## Pasangan Komutatif Supertropical Order 3 - Full Tangible

Terdapat 9 pasangan $(A, B)$ dengan $A \neq B$ yang memenuhi $A \otimes B = B \otimes A$ dimana hasil perkalian full tangible:

**1.** $L_2 \circ L_3$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (123) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (132) \\ \tau_3 = (123) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{pmatrix}$$

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**2.** $L_2 \circ L_6$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (123) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (132) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 5 \\ 5 & 5 & 6 \\ 6 & 5 & 5 \end{pmatrix}$$

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**3.** $L_2 \circ L_{11}$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (123) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (123) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 5 & 6 \\ 6 & 5 & 5 \\ 5 & 6 & 5 \end{pmatrix}$$

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**4.** $L_3 \circ L_7$

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (132) \\ \tau_3 = (123) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (123) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 5 & 6 \\ 6 & 5 & 5 \\ 5 & 6 & 5 \end{pmatrix}$$

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**5.** $L_3 \circ L_{10}$

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (132) \\ \tau_3 = (123) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (132) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 5 \\ 5 & 5 & 6 \\ 6 & 5 & 5 \end{pmatrix}$$

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**6.** $L_6 \circ L_7$

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (123) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{pmatrix}$$

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**7.** $L_6 \circ L_{10}$

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (132) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 5 & 6 \\ 6 & 5 & 5 \\ 5 & 6 & 5 \end{pmatrix}$$

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**8.** $L_7 \circ L_{11}$

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (123) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (123) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 5 \\ 5 & 5 & 6 \\ 6 & 5 & 5 \end{pmatrix}$$

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**9.** $L_{10} \circ L_{11}$

$$A = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (132) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (123) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{pmatrix}$$

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## Pasangan Komutatif Supertropical Order 3 - Contains Ghost

Terdapat 6 pasangan $(A, B)$ dengan $A \neq B$ yang memenuhi $A \otimes B = B \otimes A$ dimana hasil perkalian mengandung elemen ghost $(a^\nu)$:

**1.** $L_2 \circ L_7$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (123) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (123) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 4^\nu & 5^\nu \\ 5^\nu & 6 & 4^\nu \\ 4^\nu & 5^\nu & 6 \end{pmatrix}$$

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**2.** $L_2 \circ L_{10}$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (123) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (132) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 4^\nu & 5^\nu & 6 \\ 6 & 4^\nu & 5^\nu \\ 5^\nu & 6 & 4^\nu \end{pmatrix}$$

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**3.** $L_3 \circ L_6$

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (132) \\ \tau_3 = (123) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (132) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5^\nu & 4^\nu \\ 4^\nu & 6 & 5^\nu \\ 5^\nu & 4^\nu & 6 \end{pmatrix}$$

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**4.** $L_3 \circ L_{11}$

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3) \\ \tau_2 = (132) \\ \tau_3 = (123) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (123) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 4^\nu & 6 & 5^\nu \\ 5^\nu & 4^\nu & 6 \\ 6 & 5^\nu & 4^\nu \end{pmatrix}$$

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**5.** $L_6 \circ L_{11}$

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (132) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (123) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 4^\nu & 6 \\ 6 & 5^\nu & 4^\nu \\ 4^\nu & 6 & 5^\nu \end{pmatrix}$$

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**6.** $L_7 \circ L_{10}$

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132) \\ \tau_2 = (1)(2)(3) \\ \tau_3 = (123) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123) \\ \tau_2 = (132) \\ \tau_3 = (1)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6 & 4^\nu \\ 4^\nu & 5^\nu & 6 \\ 6 & 4^\nu & 5^\nu \end{pmatrix}$$

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## Pasangan Komutatif Supertropical Order 4 - Full Tangible

Terdapat 300 pasangan $(A, B)$ dengan $A \neq B$ yang memenuhi $A \otimes B = B \otimes A$ dimana hasil perkalian full tangible:

**1.** $L_1 \circ L_{50}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**2.** $L_1 \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**3.** $L_1 \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**4.** $L_1 \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**5.** $L_3 \circ L_{52}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**6.** $L_3 \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**7.** $L_3 \circ L_{383}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**8.** $L_3 \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**9.** $L_5 \circ L_{150}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**10.** $L_5 \circ L_{417}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**11.** $L_7 \circ L_{152}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**12.** $L_7 \circ L_{439}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**13.** $L_9 \circ L_{42}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**14.** $L_9 \circ L_{216}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**15.** $L_9 \circ L_{409}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**16.** $L_9 \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**17.** $L_{11} \circ L_{459}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**18.** $L_{18} \circ L_{34}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**19.** $L_{18} \circ L_{65}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

**20.** $L_{18} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**21.** $L_{18} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**22.** $L_{18} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

---

**23.** $L_{18} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**24.** $L_{18} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**25.** $L_{18} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**26.** $L_{25} \circ L_{90}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**27.** $L_{25} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

**28.** $L_{25} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**29.** $L_{25} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**30.** $L_{27} \circ L_{92}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**31.** $L_{27} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

**32.** $L_{27} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**33.** $L_{27} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**34.** $L_{30} \circ L_{179}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**35.** $L_{30} \circ L_{403}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**36.** $L_{32} \circ L_{177}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**37.** $L_{32} \circ L_{465}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**38.** $L_{34} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**39.** $L_{34} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**40.** $L_{34} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**41.** $L_{38} \circ L_{451}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**42.** $L_{42} \circ L_{74}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

---

**43.** $L_{42} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**44.** $L_{42} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**45.** $L_{42} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**46.** $L_{42} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**47.** $L_{42} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**48.** $L_{42} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**49.** $L_{49} \circ L_{196}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**50.** $L_{49} \circ L_{326}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**51.** $L_{50} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**52.** $L_{50} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

**53.** $L_{50} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**54.** $L_{52} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

---

**55.** $L_{52} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**56.** $L_{52} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**57.** $L_{52} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**58.** $L_{52} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

**59.** $L_{52} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**60.** $L_{52} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

---

**61.** $L_{55} \circ L_{199}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**62.** $L_{55} \circ L_{563}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**63.** $L_{57} \circ L_{105}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

**64.** $L_{57} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**65.** $L_{57} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

---

**66.** $L_{57} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**67.** $L_{59} \circ L_{499}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{array}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**68.** $L_{65} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

---

**69.** $L_{65} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**70.** $L_{65} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**71.** $L_{74} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**72.** $L_{74} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**73.** $L_{74} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**74.** $L_{81} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{array}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**75.** $L_{81} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{array}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**76.** $L_{81} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**77.** $L_{81} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**78.** $L_{86} \circ L_{526}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**79.** $L_{90} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**80.** $L_{90} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**81.** $L_{90} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**82.** $L_{90} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**83.** $L_{90} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**84.** $L_{90} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**85.** $L_{90} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**86.** $L_{91} \circ L_{238}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**87.** $L_{91} \circ L_{291}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**88.** $L_{92} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**89.** $L_{92} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**90.** $L_{92} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**91.** $L_{96} \circ L_{233}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**92.** $L_{96} \circ L_{540}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**93.** $L_{97} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**94.** $L_{97} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

---

**95.** $L_{97} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**96.** $L_{98} \circ L_{568}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**97.** $L_{99} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**98.** $L_{99} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

---

**99.** $L_{99} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

---

**100.** $L_{99} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**101.** $L_{100} \circ L_{275}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**102.** $L_{100} \circ L_{495}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**103.** $L_{104} \circ L_{279}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**104.** $L_{104} \circ L_{379}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**105.** $L_{105} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**106.** $L_{105} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**107.** $L_{105} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**108.** $L_{105} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

---

**109.** $L_{105} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**110.** $L_{105} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**111.** $L_{105} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**112.** $L_{119} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

---

**113.** $L_{119} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**114.** $L_{119} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**115.** $L_{119} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**116.** $L_{128} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**117.** $L_{128} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**118.** $L_{128} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**119.** $L_{129} \circ L_{257}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**120.** $L_{129} \circ L_{337}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**121.** $L_{133} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**122.** $L_{133} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**123.** $L_{133} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**124.** $L_{134} \circ L_{262}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**125.** $L_{134} \circ L_{518}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**126.** $L_{135} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**127.** $L_{135} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

---

**128.** $L_{135} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**129.** $L_{136} \circ L_{541}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**130.** $L_{138} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**131.** $L_{138} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**132.** $L_{138} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

---

**133.** $L_{138} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**134.** $L_{138} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**135.** $L_{138} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**136.** $L_{146} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**137.** $L_{146} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**138.** $L_{146} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**139.** $L_{148} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**140.** $L_{148} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**141.** $L_{148} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**142.** $L_{150} \circ L_{401}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**143.** $L_{152} \circ L_{467}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**144.** $L_{159} \circ L_{176}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**145.** $L_{159} \circ L_{232}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**146.** $L_{159} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**147.** $L_{159} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**148.** $L_{159} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**149.** $L_{159} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**150.** $L_{163} \circ L_{430}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**151.** $L_{167} \circ L_{186}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**152.** $L_{167} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**153.** $L_{167} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**154.** $L_{171} \circ L_{398}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**155.** $L_{176} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**156.** $L_{176} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

**157.** $L_{177} \circ L_{437}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**158.** $L_{179} \circ L_{423}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**159.** $L_{181} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**160.** $L_{181} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**161.** $L_{181} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

**162.** $L_{183} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**163.** $L_{183} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**164.** $L_{183} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**165.** $L_{186} \circ L_{202}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**166.** $L_{186} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**167.** $L_{186} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**168.** $L_{186} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**169.** $L_{186} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**170.** $L_{195} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**171.** $L_{195} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**172.** $L_{195} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**173.** $L_{196} \circ L_{382}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**174.** $L_{197} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**175.** $L_{197} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**176.** $L_{197} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**177.** $L_{198} \circ L_{328}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**178.** $L_{199} \circ L_{490}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**179.** $L_{202} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**180.** $L_{202} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**181.** $L_{216} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**182.** $L_{216} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**183.** $L_{216} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**184.** $L_{216} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**185.** $L_{216} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**186.** $L_{216} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**187.** $L_{223} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

**188.** $L_{223} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**189.** $L_{223} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**190.** $L_{223} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**191.** $L_{223} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**192.** $L_{223} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**193.** $L_{232} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**194.** $L_{232} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**195.** $L_{233} \circ L_{513}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**196.** $L_{237} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**197.** $L_{237} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**198.** $L_{238} \circ L_{340}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**199.** $L_{239} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**200.** $L_{239} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**201.** $L_{239} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**202.** $L_{240} \circ L_{289}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**203.** $L_{242} \circ L_{350}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**204.** $L_{247} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**205.** $L_{247} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**206.** $L_{256} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**207.** $L_{256} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**208.** $L_{257} \circ L_{294}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**209.** $L_{261} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**210.** $L_{261} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**211.** $L_{262} \circ L_{543}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**212.** $L_{263} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**213.** $L_{263} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**214.** $L_{263} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**215.** $L_{263} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**216.** $L_{265} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**217.** $L_{265} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

---

**218.** $L_{267} \circ L_{371}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**219.** $L_{275} \circ L_{566}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**220.** $L_{276} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**221.** $L_{276} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**222.** $L_{278} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**223.** $L_{278} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**224.** $L_{278} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**225.** $L_{278} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**226.** $L_{279} \circ L_{323}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**227.** $L_{282} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**228.** $L_{282} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**229.** $L_{290} \circ L_{383}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**230.** $L_{290} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**231.** $L_{291} \circ L_{340}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**232.** $L_{292} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**233.** $L_{292} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**234.** $L_{294} \circ L_{337}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**235.** $L_{303} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**236.** $L_{303} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**237.** $L_{311} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**238.** $L_{311} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**239.** $L_{311} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**240.** $L_{311} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**241.** $L_{320} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**242.** $L_{320} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**243.** $L_{320} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**244.** $L_{320} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**245.** $L_{323} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**246.** $L_{325} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**247.** $L_{326} \circ L_{382}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**248.** $L_{327} \circ L_{341}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**249.** $L_{327} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**250.** $L_{330} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**251.** $L_{330} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**252.** $L_{339} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**253.** $L_{339} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**254.** $L_{341} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**255.** $L_{341} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**256.** $L_{341} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**257.** $L_{346} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**258.** $L_{346} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**259.** $L_{360} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**260.** $L_{367} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**261.** $L_{376} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**262.** $L_{376} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**263.** $L_{381} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**264.** $L_{381} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**265.** $L_{383} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**266.** $L_{383} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**267.** $L_{383} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**268.** $L_{391} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**269.** $L_{391} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**270.** $L_{400} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**271.** $L_{401} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**272.** $L_{403} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**273.** $L_{405} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**274.** $L_{407} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**275.** $L_{409} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**276.** $L_{409} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**277.** $L_{420} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**278.** $L_{422} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**279.** $L_{426} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**280.** $L_{434} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

**281.** $L_{436} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**282.** $L_{437} \circ L_{465}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**283.** $L_{439} \circ L_{467}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**284.** $L_{447} \circ L_{503}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**285.** $L_{447} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**286.** $L_{455} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**287.** $L_{464} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**288.** $L_{469} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**289.** $L_{471} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**290.** $L_{474} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**291.** $L_{474} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**292.** $L_{490} \circ L_{563}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**293.** $L_{494} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**294.** $L_{495} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**295.** $L_{513} \circ L_{540}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**296.** $L_{518} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**297.** $L_{519} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**298.** $L_{535} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

**299.** $L_{544} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

---

**300.** $L_{551} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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## Pasangan Komutatif Supertropical Order 4 - Contains Ghost

Terdapat 2508 pasangan $(A, B)$ dengan $A \neq B$ yang memenuhi $A \otimes B = B \otimes A$ dimana hasil perkalian mengandung elemen ghost $(a^\nu)$:

**1.** $L_1 \circ L_{27}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

---

**2.** $L_1 \circ L_{92}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**3.** $L_1 \circ L_{94}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**4.** $L_1 \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

---

**5.** $L_1 \circ L_{131}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**6.** $L_1 \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**7.** $L_1 \circ L_{148}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**8.** $L_1 \circ L_{181}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**9.** $L_1 \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**10.** $L_1 \circ L_{235}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**11.** $L_1 \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**12.** $L_1 \circ L_{259}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**13.** $L_1 \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**14.** $L_1 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**15.** $L_1 \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**16.** $L_1 \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**17.** $L_1 \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**18.** $L_1 \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**19.** $L_1 \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**20.** $L_1 \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**21.** $L_1 \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**22.** $L_1 \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**23.** $L_1 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**24.** $L_1 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**25.** $L_1 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

---

**26.** $L_1 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**27.** $L_1 \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**28.** $L_2 \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

---

**29.** $L_2 \circ L_{131}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**30.** $L_2 \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**31.** $L_2 \circ L_{182}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

---

**32.** $L_2 \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**33.** $L_2 \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**34.** $L_2 \circ L_{259}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**35.** $L_2 \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**36.** $L_2 \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**37.** $L_2 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**38.** $L_2 \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**39.** $L_2 \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**40.** $L_2 \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**41.** $L_2 \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**42.** $L_2 \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

---

**43.** $L_2 \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**44.** $L_2 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

**45.** $L_2 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

**46.** $L_2 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

---

**47.** $L_2 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

---

**48.** $L_2 \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**49.** $L_3 \circ L_{25}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

---

**50.** $L_3 \circ L_{90}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

---

**51.** $L_3 \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**52.** $L_3 \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**53.** $L_3 \circ L_{146}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

---

**54.** $L_3 \circ L_{183}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

---

**55.** $L_3 \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

---

**56.** $L_3 \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

---

**57.** $L_3 \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**58.** $L_3 \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**59.** $L_3 \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**60.** $L_3 \circ L_{341}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**61.** $L_3 \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

---

**62.** $L_3 \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

---

**63.** $L_3 \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

---

**64.** $L_3 \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**65.** $L_3 \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

---

**66.** $L_3 \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**67.** $L_3 \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**68.** $L_4 \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**69.** $L_4 \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**70.** $L_4 \circ L_{184}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

---

**71.** $L_4 \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

---

**72.** $L_4 \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

---

**73.** $L_4 \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**74.** $L_4 \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

**75.** $L_4 \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**76.** $L_4 \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**77.** $L_4 \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

---

**78.** $L_4 \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

---

**79.** $L_4 \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

---

**80.** $L_4 \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**81.** $L_4 \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**82.** $L_4 \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**83.** $L_4 \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**84.** $L_4 \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

---

**85.** $L_5 \circ L_{181}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**86.** $L_5 \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

---

**87.** $L_5 \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**88.** $L_5 \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

---

**89.** $L_5 \circ L_{401}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**90.** $L_5 \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

---

**91.** $L_5 \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

---

**92.** $L_5 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 8 & 7 & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**93.** $L_5 \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**94.** $L_7 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**95.** $L_7 \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**96.** $L_7 \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**97.** $L_7 \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**98.** $L_7 \circ L_{467}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**99.** $L_7 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**100.** $L_7 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**101.** $L_7 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**102.** $L_7 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**103.** $L_9 \circ L_{57}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5^\nu \\ 7 & 8 & 5^\nu & 7 \\ 7 & 5^\nu & 8 & 7 \\ 5^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**104.** $L_9 \circ L_{74}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**105.** $L_9 \circ L_{77}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

---

**106.** $L_9 \circ L_{105}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 6 \\ 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 6 & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

---

**107.** $L_9 \circ L_{123}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**108.** $L_9 \circ L_{128}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**109.** $L_9 \circ L_{159}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6^\nu \\ 6 & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 6 \\ 6^\nu & 6 & 7^\nu & 8 \end{pmatrix}$$

---

**110.** $L_9 \circ L_{176}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5^\nu & 7 \\ 5^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 5^\nu \\ 7 & 5^\nu & 8 & 7 \end{pmatrix}$$

---

**111.** $L_9 \circ L_{227}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**112.** $L_9 \circ L_{232}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

---

**113.** $L_9 \circ L_{254}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**114.** $L_9 \circ L_{256}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**115.** $L_9 \circ L_{265}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6^\nu \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 6^\nu & 8 & 6 & 7^\nu \end{pmatrix}$$

---

**116.** $L_9 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

---

**117.** $L_9 \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**118.** $L_9 \circ L_{320}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6^\nu & 7^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \\ 7^\nu & 6^\nu & 8 & 6 \end{pmatrix}$$

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**119.** $L_9 \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**120.** $L_9 \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**121.** $L_9 \circ L_{376}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 6 & 7^\nu \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 7^\nu & 6 & 8 & 6^\nu \end{pmatrix}$$

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**122.** $L_9 \circ L_{400}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 8 & 7 \\ 8 & 7 & 7 & 5^\nu \\ 5^\nu & 7 & 7 & 8 \\ 7 & 8 & 5^\nu & 7 \end{pmatrix}$$

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**123.** $L_9 \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 7^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 6 \\ 6 & 8 & 6^\nu & 7^\nu \\ 8 & 7^\nu & 6 & 6^\nu \end{pmatrix}$$

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**124.** $L_9 \circ L_{464}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 7^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \\ 6^\nu & 8 & 6 & 7^\nu \\ 8 & 7^\nu & 6^\nu & 6 \end{pmatrix}$$

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**125.** $L_9 \circ L_{503}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**126.** $L_9 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**127.** $L_9 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**128.** $L_9 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**129.** $L_9 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**130.** $L_{11} \circ L_{156}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**131.** $L_{11} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**132.** $L_{11} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**133.** $L_{11} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**134.** $L_{11} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**135.** $L_{11} \circ L_{444}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**136.** $L_{11} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**137.** $L_{11} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**138.** $L_{11} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**139.** $L_{11} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**140.** $L_{15} \circ L_{156}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**141.** $L_{15} \circ L_{444}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**142.** $L_{18} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \end{pmatrix}$$

---

**143.** $L_{18} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**144.** $L_{18} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**145.** $L_{18} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \end{pmatrix}$$

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**146.** $L_{18} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**147.** $L_{18} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**148.** $L_{18} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**149.** $L_{18} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**150.** $L_{18} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**151.** $L_{18} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**152.** $L_{18} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**153.** $L_{18} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**154.** $L_{18} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**155.** $L_{18} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \end{pmatrix}$$

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**156.** $L_{18} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**157.** $L_{20} \circ L_{24}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**158.** $L_{20} \circ L_{386}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**159.** $L_{20} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**160.** $L_{20} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**161.** $L_{24} \circ L_{161}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**162.** $L_{24} \circ L_{163}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**163.** $L_{24} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**164.** $L_{24} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**165.** $L_{24} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**166.** $L_{24} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**167.** $L_{24} \circ L_{386}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**168.** $L_{24} \circ L_{393}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**169.** $L_{24} \circ L_{416}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**170.** $L_{24} \circ L_{430}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**171.** $L_{24} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**172.** $L_{24} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**173.** $L_{24} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**174.** $L_{24} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**175.** $L_{25} \circ L_{52}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**176.** $L_{25} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**177.** $L_{25} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**178.** $L_{25} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \end{pmatrix}$$

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**179.** $L_{25} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**180.** $L_{25} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**181.** $L_{25} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**182.** $L_{25} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**183.** $L_{25} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**184.** $L_{25} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**185.** $L_{25} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

---

**186.** $L_{25} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**187.** $L_{25} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**188.** $L_{25} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**189.** $L_{25} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**190.** $L_{25} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**191.** $L_{25} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**192.** $L_{25} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**193.** $L_{26} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**194.** $L_{26} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**195.** $L_{26} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**196.** $L_{26} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**197.** $L_{26} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**198.** $L_{26} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**199.** $L_{26} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**200.** $L_{26} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**201.** $L_{26} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**202.** $L_{26} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**203.** $L_{26} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**204.** $L_{26} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**205.** $L_{26} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**206.** $L_{26} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{array}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**207.** $L_{26} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**208.** $L_{26} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**209.** $L_{26} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**210.** $L_{27} \circ L_{50}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**211.** $L_{27} \circ L_{53}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**212.** $L_{27} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**213.** $L_{27} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**214.** $L_{27} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**215.** $L_{27} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**216.** $L_{27} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**217.** $L_{27} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**218.** $L_{27} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**219.** $L_{27} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**220.** $L_{27} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**221.** $L_{27} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**222.** $L_{27} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**223.** $L_{27} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**224.** $L_{27} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**225.** $L_{27} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**226.** $L_{27} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**227.** $L_{27} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**228.** $L_{27} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**229.** $L_{27} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**230.** $L_{27} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**231.** $L_{27} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**232.** $L_{27} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**233.** $L_{27} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**234.** $L_{27} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**235.** $L_{27} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**236.** $L_{28} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**237.** $L_{28} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**238.** $L_{28} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**239.** $L_{28} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**240.** $L_{28} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**241.** $L_{28} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**242.** $L_{28} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**243.** $L_{28} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**244.** $L_{28} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**245.** $L_{28} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**246.** $L_{28} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**247.** $L_{28} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**248.** $L_{28} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**249.** $L_{28} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**250.** $L_{28} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**251.** $L_{28} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**252.** $L_{28} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**253.** $L_{28} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**254.** $L_{28} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**255.** $L_{28} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**256.** $L_{28} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**257.** $L_{30} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**258.** $L_{30} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**259.** $L_{30} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**260.** $L_{30} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**261.** $L_{30} \circ L_{423}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**262.** $L_{30} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**263.** $L_{30} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**264.** $L_{30} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**265.** $L_{30} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**266.** $L_{32} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**267.** $L_{32} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**268.** $L_{32} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**269.** $L_{32} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**270.** $L_{32} \circ L_{437}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**271.** $L_{32} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**272.** $L_{32} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**273.** $L_{32} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**274.** $L_{32} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**275.** $L_{34} \circ L_{65}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**276.** $L_{34} \circ L_{70}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**277.** $L_{34} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \end{pmatrix}$$

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**278.** $L_{34} \circ L_{115}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**279.** $L_{34} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**280.** $L_{34} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \end{pmatrix}$$

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**281.** $L_{34} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \end{pmatrix}$$

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**282.** $L_{34} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \end{pmatrix}$$

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**283.** $L_{34} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**284.** $L_{34} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**285.** $L_{34} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**286.** $L_{34} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**287.** $L_{34} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**288.** $L_{34} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \end{pmatrix}$$

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**289.** $L_{34} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**290.** $L_{34} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**291.** $L_{34} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 6 & 6^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \end{pmatrix}$$

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**292.** $L_{34} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**293.** $L_{34} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**294.** $L_{34} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**295.** $L_{34} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 6 & 6^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 6 \end{pmatrix}$$

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**296.** $L_{34} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**297.** $L_{34} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**298.** $L_{34} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**299.** $L_{34} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**300.** $L_{34} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**301.** $L_{34} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**302.** $L_{38} \circ L_{190}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**303.** $L_{38} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**304.** $L_{38} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**305.** $L_{38} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**306.** $L_{38} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**307.** $L_{38} \circ L_{478}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**308.** $L_{38} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**309.** $L_{38} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**310.** $L_{38} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**311.** $L_{38} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**312.** $L_{40} \circ L_{190}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

---

**313.** $L_{40} \circ L_{478}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**314.** $L_{42} \circ L_{57}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 6 \\ 6^\nu & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6^\nu \\ 6 & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

---

**315.** $L_{42} \circ L_{105}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**316.** $L_{42} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**317.** $L_{42} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**318.** $L_{42} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 6 & 8 \\ 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 8 & 6 & 7^\nu & 6^\nu \end{pmatrix}$$

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**319.** $L_{42} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**320.** $L_{42} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**321.** $L_{42} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**322.** $L_{42} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**323.** $L_{42} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7^\nu & 6 & 8 \end{pmatrix}$$

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**324.** $L_{42} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \end{pmatrix}$$

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**325.** $L_{42} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**326.** $L_{42} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 8 & 7^\nu \\ 8 & 6 & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 6 & 8 \\ 7^\nu & 8 & 6^\nu & 6 \end{pmatrix}$$

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**327.** $L_{42} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**328.** $L_{42} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**329.** $L_{45} \circ L_{48}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**330.** $L_{45} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**331.** $L_{45} \circ L_{171}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**332.** $L_{45} \circ L_{174}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**333.** $L_{45} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**334.** $L_{45} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**335.** $L_{45} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**336.** $L_{45} \circ L_{390}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**337.** $L_{45} \circ L_{398}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**338.** $L_{45} \circ L_{411}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**339.** $L_{45} \circ L_{431}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**340.** $L_{45} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**341.** $L_{45} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**342.** $L_{45} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

**343.** $L_{45} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**344.** $L_{48} \circ L_{411}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**345.** $L_{48} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**346.** $L_{48} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**347.** $L_{49} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**348.** $L_{49} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**349.** $L_{49} \circ L_{382}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**350.** $L_{49} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**351.** $L_{49} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**352.** $L_{49} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**353.** $L_{49} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**354.** $L_{49} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**355.** $L_{49} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**356.** $L_{50} \circ L_{92}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**357.** $L_{50} \circ L_{94}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**358.** $L_{50} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**359.** $L_{50} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**360.** $L_{50} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**361.** $L_{50} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**362.** $L_{50} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**363.** $L_{50} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**364.** $L_{50} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**365.** $L_{50} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**366.** $L_{50} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{array}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**367.** $L_{50} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**368.** $L_{50} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**369.** $L_{50} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**370.** $L_{50} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**371.** $L_{50} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**372.** $L_{50} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**373.** $L_{50} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**374.** $L_{50} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**375.** $L_{50} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**376.** $L_{50} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**377.** $L_{50} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**378.** $L_{50} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**379.** $L_{50} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**380.** $L_{50} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**381.** $L_{50} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \\ 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**382.** $L_{51} \circ L_{54}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**383.** $L_{51} \circ L_{194}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**384.** $L_{51} \circ L_{198}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**385.** $L_{51} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**386.** $L_{51} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**387.** $L_{51} \circ L_{328}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**388.** $L_{51} \circ L_{329}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**389.** $L_{51} \circ L_{368}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**390.** $L_{51} \circ L_{384}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**391.** $L_{51} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**392.** $L_{51} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**393.** $L_{51} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**394.** $L_{51} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**395.** $L_{51} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**396.** $L_{51} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**397.** $L_{52} \circ L_{90}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**398.** $L_{52} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \end{pmatrix}$$

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**399.** $L_{52} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**400.** $L_{52} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \end{pmatrix}$$

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**401.** $L_{52} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**402.** $L_{52} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**403.** $L_{52} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**404.** $L_{52} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**405.** $L_{52} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**406.** $L_{52} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**407.** $L_{52} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**408.** $L_{52} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**409.** $L_{52} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**410.** $L_{52} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \\ 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \end{pmatrix}$$

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**411.** $L_{53} \circ L_{92}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**412.** $L_{53} \circ L_{94}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**413.** $L_{53} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**414.** $L_{53} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**415.** $L_{53} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**416.** $L_{53} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**417.** $L_{53} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**418.** $L_{53} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**419.** $L_{53} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**420.** $L_{53} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**421.** $L_{53} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**422.** $L_{53} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**423.** $L_{53} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**424.** $L_{53} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**425.** $L_{53} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**426.** $L_{53} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**427.** $L_{53} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**428.** $L_{53} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**429.** $L_{53} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**430.** $L_{53} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 8 & 6 & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**431.** $L_{54} \circ L_{384}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**432.** $L_{54} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**433.** $L_{54} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**434.** $L_{55} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**435.** $L_{55} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**436.** $L_{55} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**437.** $L_{55} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**438.** $L_{55} \circ L_{490}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**439.** $L_{55} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**440.** $L_{55} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**441.** $L_{55} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**442.** $L_{55} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**443.** $L_{57} \circ L_{74}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**444.** $L_{57} \circ L_{77}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**445.** $L_{57} \circ L_{123}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**446.** $L_{57} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**447.** $L_{57} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 6 & 8 & 7^\nu \end{pmatrix}$$

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**448.** $L_{57} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7^\nu & 6 & 8 \end{pmatrix}$$

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**449.** $L_{57} \circ L_{227}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**450.** $L_{57} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**451.** $L_{57} \circ L_{254}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**452.** $L_{57} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**453.** $L_{57} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 8 & 7 \\ 8 & 7 & 7 & 5^\nu \\ 5^\nu & 7 & 7 & 8 \\ 7 & 8 & 5^\nu & 7 \end{pmatrix}$$

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**454.** $L_{57} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**455.** $L_{57} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**456.** $L_{57} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**457.** $L_{57} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**458.** $L_{57} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5^\nu & 7 \\ 5^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 5^\nu \\ 7 & 5^\nu & 8 & 7 \end{pmatrix}$$

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**459.** $L_{57} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 8 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 8 & 6 & 6^\nu \end{pmatrix}$$

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**460.** $L_{57} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 8 & 7^\nu \\ 8 & 6 & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 6 & 8 \\ 7^\nu & 8 & 6^\nu & 6 \end{pmatrix}$$

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**461.** $L_{57} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**462.** $L_{57} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6^\nu & 8 \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 6 \end{pmatrix}$$

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**463.** $L_{57} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**464.** $L_{57} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**465.** $L_{57} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**466.** $L_{57} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**467.** $L_{57} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 6 & 8 \\ 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 8 & 6 & 7^\nu & 6^\nu \end{pmatrix}$$

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**468.** $L_{59} \circ L_{214}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**469.** $L_{59} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**470.** $L_{59} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**471.** $L_{59} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**472.** $L_{59} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**473.** $L_{59} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**474.** $L_{59} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**475.** $L_{59} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**476.** $L_{59} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**477.** $L_{59} \circ L_{555}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**478.** $L_{60} \circ L_{214}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**479.** $L_{60} \circ L_{555}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**480.** $L_{65} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**481.** $L_{65} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**482.** $L_{65} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**483.** $L_{65} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**484.** $L_{65} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**485.** $L_{65} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**486.** $L_{65} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**487.** $L_{65} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**488.** $L_{65} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**489.** $L_{65} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**490.** $L_{65} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**491.** $L_{65} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**492.** $L_{65} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**493.** $L_{65} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**494.** $L_{65} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**495.** $L_{65} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**496.** $L_{65} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**497.** $L_{65} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**498.** $L_{70} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**499.** $L_{70} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**500.** $L_{70} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**501.** $L_{70} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**502.** $L_{70} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**503.** $L_{70} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**504.** $L_{70} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**505.** $L_{70} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**506.** $L_{70} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**507.** $L_{70} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**508.** $L_{70} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**509.** $L_{70} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**510.** $L_{70} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**511.** $L_{70} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**512.** $L_{70} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**513.** $L_{70} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**514.** $L_{74} \circ L_{105}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**515.** $L_{74} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**516.** $L_{74} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**517.** $L_{74} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**518.** $L_{74} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**519.** $L_{74} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 6^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \end{pmatrix}$$

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**520.** $L_{74} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**521.** $L_{74} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**522.** $L_{74} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**523.** $L_{74} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**524.** $L_{74} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**525.** $L_{74} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**526.** $L_{74} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**527.** $L_{74} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**528.** $L_{74} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**529.** $L_{74} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**530.** $L_{74} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**531.** $L_{77} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**532.** $L_{77} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**533.** $L_{77} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**534.** $L_{77} \circ L_{254}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**535.** $L_{77} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**536.** $L_{77} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**537.** $L_{77} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**538.** $L_{77} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**539.** $L_{77} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**540.** $L_{77} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**541.** $L_{77} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**542.** $L_{77} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**543.** $L_{77} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**544.** $L_{77} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**545.** $L_{77} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**546.** $L_{81} \circ L_{115}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**547.** $L_{81} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**548.** $L_{81} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 6 \\ 6 & 7^\nu & 8 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 8 & 6^\nu & 6 & 7^\nu \end{pmatrix}$$

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**549.** $L_{81} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**550.** $L_{81} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**551.** $L_{81} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**552.** $L_{81} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**553.** $L_{81} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**554.** $L_{81} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**555.** $L_{81} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**556.** $L_{81} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**557.** $L_{81} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \end{pmatrix}$$

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**558.** $L_{81} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**559.** $L_{81} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**560.** $L_{81} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \end{pmatrix}$$

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**561.** $L_{81} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 7^\nu & 8 \\ 8 & 6^\nu & 6 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 6 & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**562.** $L_{81} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**563.** $L_{81} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**564.** $L_{81} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**565.** $L_{81} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 8 & 6^\nu & 6 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \end{pmatrix}$$

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**566.** $L_{81} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \end{pmatrix}$$

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**567.** $L_{81} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**568.** $L_{81} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**569.** $L_{85} \circ L_{219}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**570.** $L_{85} \circ L_{530}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**571.** $L_{86} \circ L_{219}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

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**572.** $L_{86} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**573.** $L_{86} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**574.** $L_{86} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**575.** $L_{86} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**576.** $L_{86} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{array}{c} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{array}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**577.** $L_{86} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**578.** $L_{86} \circ L_{530}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**579.** $L_{86} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**580.** $L_{86} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**581.** $L_{89} \circ L_{93}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**582.** $L_{89} \circ L_{236}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**583.** $L_{89} \circ L_{240}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**584.** $L_{89} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**585.** $L_{89} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**586.** $L_{89} \circ L_{289}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**587.** $L_{89} \circ L_{304}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**588.** $L_{89} \circ L_{342}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**589.** $L_{89} \circ L_{359}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**590.** $L_{89} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**591.** $L_{89} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**592.** $L_{89} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

**593.** $L_{89} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**594.** $L_{89} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**595.** $L_{89} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**596.** $L_{90} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \end{pmatrix}$$

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**597.** $L_{90} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**598.** $L_{90} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**599.** $L_{90} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**600.** $L_{90} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**601.** $L_{90} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \end{pmatrix}$$

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**602.** $L_{90} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \end{pmatrix}$$

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**603.** $L_{90} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**604.** $L_{90} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**605.** $L_{90} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**606.** $L_{90} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**607.** $L_{90} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**608.** $L_{90} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \\ 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \end{pmatrix}$$

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**609.** $L_{91} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**610.** $L_{91} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**611.** $L_{91} \circ L_{340}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**612.** $L_{91} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**613.** $L_{91} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**614.** $L_{91} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**615.** $L_{91} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**616.** $L_{91} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**617.** $L_{91} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**618.** $L_{92} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**619.** $L_{92} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**620.** $L_{92} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**621.** $L_{92} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**622.** $L_{92} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**623.** $L_{92} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**624.** $L_{92} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**625.** $L_{92} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**626.** $L_{92} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**627.** $L_{92} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**628.** $L_{92} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**629.** $L_{92} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**630.** $L_{92} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**631.** $L_{92} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**632.** $L_{92} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**633.** $L_{92} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**634.** $L_{92} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**635.** $L_{92} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**636.** $L_{92} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**637.** $L_{92} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**638.** $L_{92} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**639.** $L_{92} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**640.** $L_{92} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**641.** $L_{92} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**642.** $L_{93} \circ L_{342}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**643.** $L_{93} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**644.** $L_{93} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**645.** $L_{94} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**646.** $L_{94} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**647.** $L_{94} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**648.** $L_{94} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**649.** $L_{94} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**650.** $L_{94} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**651.** $L_{94} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**652.** $L_{94} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**653.** $L_{94} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**654.** $L_{94} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**655.** $L_{94} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**656.** $L_{94} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**657.** $L_{94} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**658.** $L_{94} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**659.** $L_{94} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**660.** $L_{94} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**661.** $L_{94} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**662.** $L_{94} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**663.** $L_{96} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**664.** $L_{96} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**665.** $L_{96} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**666.** $L_{96} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**667.** $L_{96} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**668.** $L_{96} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**669.** $L_{96} \circ L_{513}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**670.** $L_{96} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**671.** $L_{96} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**672.** $L_{97} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \end{pmatrix}$$

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**673.** $L_{97} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**674.** $L_{97} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**675.** $L_{97} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**676.** $L_{97} \circ L_{184}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**677.** $L_{97} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \\ 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \end{pmatrix}$$

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**678.** $L_{97} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 6 & 8 \\ 6^\nu & 7^\nu & 8 & 6 \\ 8 & 6 & 7^\nu & 6^\nu \\ 6 & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**679.** $L_{97} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \end{pmatrix}$$

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**680.** $L_{97} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 6 \\ 7^\nu & 6^\nu & 6 & 8 \\ 6 & 8 & 6^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6^\nu \end{pmatrix}$$

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**681.** $L_{97} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \\ 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**682.** $L_{97} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \\ 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \end{pmatrix}$$

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**683.** $L_{97} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**684.** $L_{97} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**685.** $L_{97} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**686.** $L_{97} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**687.** $L_{97} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**688.** $L_{97} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**689.** $L_{97} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**690.** $L_{97} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**691.** $L_{97} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**692.** $L_{97} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**693.** $L_{97} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 6^\nu & 7^\nu & 8 & 6 \end{pmatrix}$$

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**694.** $L_{98} \circ L_{277}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**695.** $L_{98} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**696.** $L_{98} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**697.** $L_{98} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**698.** $L_{98} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**699.** $L_{98} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**700.** $L_{98} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**701.** $L_{98} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**702.** $L_{98} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**703.** $L_{98} \circ L_{493}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**704.** $L_{99} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**705.** $L_{99} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**706.** $L_{99} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**707.** $L_{99} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**708.** $L_{99} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**709.** $L_{99} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**710.** $L_{99} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**711.** $L_{99} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**712.** $L_{99} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**713.** $L_{99} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \\ 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**714.** $L_{99} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**715.** $L_{99} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**716.** $L_{99} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**717.** $L_{99} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**718.** $L_{99} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**719.** $L_{99} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**720.** $L_{99} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**721.** $L_{99} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**722.** $L_{99} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**723.** $L_{99} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**724.** $L_{99} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**725.** $L_{100} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**726.** $L_{100} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**727.** $L_{100} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**728.** $L_{100} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**729.** $L_{100} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**730.** $L_{100} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**731.** $L_{100} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**732.** $L_{100} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**733.** $L_{100} \circ L_{566}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**734.** $L_{101} \circ L_{277}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**735.** $L_{101} \circ L_{493}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**736.** $L_{102} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**737.** $L_{102} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**738.** $L_{102} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**739.** $L_{102} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**740.** $L_{102} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**741.** $L_{102} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**742.** $L_{102} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**743.** $L_{102} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**744.** $L_{102} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**745.** $L_{102} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**746.** $L_{102} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**747.** $L_{102} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**748.** $L_{102} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**749.** $L_{102} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**750.** $L_{102} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**751.** $L_{102} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**752.** $L_{102} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**753.** $L_{102} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**754.** $L_{104} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**755.** $L_{104} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**756.** $L_{104} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**757.** $L_{104} \circ L_{323}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**758.** $L_{104} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**759.** $L_{104} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**760.** $L_{104} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**761.** $L_{104} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**762.** $L_{104} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**763.** $L_{105} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**764.** $L_{105} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**765.** $L_{105} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**766.** $L_{105} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 7^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 6 \\ 6 & 8 & 6^\nu & 7^\nu \\ 8 & 7^\nu & 6 & 6^\nu \end{pmatrix}$$

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**767.** $L_{105} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**768.** $L_{105} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 5^\nu & 6^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \end{pmatrix}$$

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**769.** $L_{105} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6^\nu \\ 6 & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 6 \\ 6^\nu & 6 & 7^\nu & 8 \end{pmatrix}$$

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**770.** $L_{105} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**771.** $L_{105} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**772.** $L_{105} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6^\nu & 7^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \\ 7^\nu & 6^\nu & 8 & 6 \end{pmatrix}$$

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**773.** $L_{105} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**774.** $L_{105} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**775.** $L_{107} \circ L_{108}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**776.** $L_{107} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**777.** $L_{107} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**778.** $L_{107} \circ L_{267}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**779.** $L_{107} \circ L_{268}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**780.** $L_{107} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**781.** $L_{107} \circ L_{313}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**782.** $L_{107} \circ L_{315}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**783.** $L_{107} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**784.** $L_{107} \circ L_{369}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**785.** $L_{107} \circ L_{371}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**786.** $L_{107} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**787.** $L_{107} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**788.** $L_{107} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**789.** $L_{107} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**790.** $L_{108} \circ L_{315}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**791.** $L_{108} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**792.** $L_{108} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**793.** $L_{115} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**794.** $L_{115} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**795.** $L_{115} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**796.** $L_{115} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**797.** $L_{115} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**798.** $L_{115} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**799.** $L_{115} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**800.** $L_{115} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**801.** $L_{115} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**802.** $L_{115} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**803.** $L_{115} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**804.** $L_{115} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**805.** $L_{115} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**806.** $L_{115} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**807.** $L_{115} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**808.** $L_{119} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**809.** $L_{119} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \end{pmatrix}$$

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**810.** $L_{119} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**811.** $L_{119} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**812.** $L_{119} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**813.** $L_{119} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**814.** $L_{119} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**815.** $L_{119} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**816.** $L_{119} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**817.** $L_{119} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**818.** $L_{119} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**819.** $L_{119} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**820.** $L_{119} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**821.** $L_{119} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**822.** $L_{119} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**823.** $L_{123} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**824.** $L_{123} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**825.** $L_{123} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**826.** $L_{123} \circ L_{227}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

---

**827.** $L_{123} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

---

**828.** $L_{123} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**829.** $L_{123} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**830.** $L_{123} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

---

**831.** $L_{123} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**832.** $L_{123} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**833.** $L_{123} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**834.** $L_{123} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**835.** $L_{123} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**836.** $L_{123} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**837.** $L_{123} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**838.** $L_{128} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

---

**839.** $L_{128} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**840.** $L_{128} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 5^\nu & 7^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \end{pmatrix}$$

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**841.** $L_{128} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**842.** $L_{128} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**843.** $L_{128} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**844.** $L_{128} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**845.** $L_{128} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**846.** $L_{128} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

---

**847.** $L_{128} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**848.** $L_{128} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**849.** $L_{128} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**850.** $L_{128} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**851.** $L_{128} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**852.** $L_{128} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**853.** $L_{129} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**854.** $L_{129} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**855.** $L_{129} \circ L_{294}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**856.** $L_{129} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**857.** $L_{129} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**858.** $L_{129} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**859.** $L_{129} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**860.** $L_{129} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**861.** $L_{129} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**862.** $L_{131} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**863.** $L_{131} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**864.** $L_{131} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**865.** $L_{131} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**866.** $L_{131} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**867.** $L_{131} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**868.** $L_{131} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**869.** $L_{131} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**870.** $L_{131} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**871.** $L_{131} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**872.** $L_{131} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**873.** $L_{131} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**874.** $L_{131} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**875.** $L_{131} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6 & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**876.** $L_{131} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**877.** $L_{131} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**878.** $L_{131} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**879.** $L_{131} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**880.** $L_{132} \circ L_{264}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**881.** $L_{132} \circ L_{520}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**882.** $L_{133} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**883.** $L_{133} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**884.** $L_{133} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**885.** $L_{133} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**886.** $L_{133} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**887.** $L_{133} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**888.** $L_{133} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**889.** $L_{133} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**890.** $L_{133} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**891.** $L_{133} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**892.** $L_{133} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**893.** $L_{133} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**894.** $L_{133} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**895.** $L_{133} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**896.** $L_{133} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**897.** $L_{133} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**898.** $L_{133} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**899.** $L_{133} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**900.** $L_{133} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**901.** $L_{133} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**902.** $L_{133} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**903.** $L_{134} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**904.** $L_{134} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**905.** $L_{134} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**906.** $L_{134} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**907.** $L_{134} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**908.** $L_{134} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**909.** $L_{134} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**910.** $L_{134} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**911.** $L_{134} \circ L_{543}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**912.** $L_{135} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**913.** $L_{135} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**914.** $L_{135} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**915.** $L_{135} \circ L_{184}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**916.** $L_{135} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 6 \\ 6^\nu & 7^\nu & 6 & 8 \\ 6 & 8 & 7^\nu & 6^\nu \\ 8 & 6 & 6^\nu & 7^\nu \end{pmatrix}$$

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**917.** $L_{135} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \\ 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**918.** $L_{135} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**919.** $L_{135} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \\ 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \end{pmatrix}$$

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**920.** $L_{135} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 6 & 8 \\ 7^\nu & 6^\nu & 8 & 6 \\ 8 & 6 & 6^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**921.** $L_{135} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \\ 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \end{pmatrix}$$

---

**922.** $L_{135} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**923.** $L_{135} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**924.** $L_{135} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**925.** $L_{135} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**926.** $L_{135} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**927.** $L_{135} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**928.** $L_{135} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**929.** $L_{135} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**930.** $L_{135} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \end{pmatrix}$$

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**931.** $L_{135} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6^\nu \\ 8 & 6 & 6^\nu & 7^\nu \\ 6^\nu & 7^\nu & 6 & 8 \\ 7^\nu & 6^\nu & 8 & 6 \end{pmatrix}$$

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**932.** $L_{135} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**933.** $L_{136} \circ L_{264}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**934.** $L_{136} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**935.** $L_{136} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**936.** $L_{136} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**937.** $L_{136} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**938.** $L_{136} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**939.** $L_{136} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**940.** $L_{136} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**941.** $L_{136} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**942.** $L_{136} \circ L_{520}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**943.** $L_{138} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**944.** $L_{138} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**945.** $L_{138} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**946.** $L_{138} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**947.** $L_{138} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \end{pmatrix}$$

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**948.** $L_{138} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**949.** $L_{138} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \end{pmatrix}$$

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**950.** $L_{138} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**951.** $L_{138} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \end{pmatrix}$$

---

**952.** $L_{138} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**953.** $L_{138} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**954.** $L_{138} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**955.** $L_{141} \circ L_{142}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**956.** $L_{141} \circ L_{307}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**957.** $L_{141} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**958.** $L_{141} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**959.** $L_{142} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**960.** $L_{142} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**961.** $L_{142} \circ L_{241}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**962.** $L_{142} \circ L_{242}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**963.** $L_{142} \circ L_{307}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**964.** $L_{142} \circ L_{309}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**965.** $L_{142} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**966.** $L_{142} \circ L_{348}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

---

**967.** $L_{142} \circ L_{350}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**968.** $L_{142} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**969.** $L_{142} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**970.** $L_{142} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**971.** $L_{142} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

---

**972.** $L_{142} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**973.** $L_{145} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**974.** $L_{145} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**975.** $L_{145} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

**976.** $L_{145} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**977.** $L_{145} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**978.** $L_{145} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**979.** $L_{145} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**980.** $L_{145} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**981.** $L_{145} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**982.** $L_{145} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

---

**983.** $L_{145} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**984.** $L_{145} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**985.** $L_{145} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**986.** $L_{145} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**987.** $L_{146} \circ L_{183}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**988.** $L_{146} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**989.** $L_{146} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**990.** $L_{146} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**991.** $L_{146} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**992.** $L_{146} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**993.** $L_{146} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**994.** $L_{146} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**995.** $L_{146} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**996.** $L_{146} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**997.** $L_{146} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**998.** $L_{146} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**999.** $L_{146} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1000.** $L_{146} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1001.** $L_{147} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1002.** $L_{147} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1003.** $L_{147} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1004.** $L_{147} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1005.** $L_{147} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**1006.** $L_{147} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1007.** $L_{147} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1008.** $L_{147} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**1009.** $L_{147} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**1010.** $L_{147} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**1011.** $L_{147} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**1012.** $L_{147} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**1013.** $L_{147} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1014.** $L_{147} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1015.** $L_{147} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1016.** $L_{147} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1017.** $L_{147} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1018.** $L_{148} \circ L_{159}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 5^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 5^\nu & 7^\nu & 8 \end{pmatrix}$$

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**1019.** $L_{148} \circ L_{181}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**1020.** $L_{148} \circ L_{193}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1021.** $L_{148} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1022.** $L_{148} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1023.** $L_{148} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1024.** $L_{148} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1025.** $L_{148} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**1026.** $L_{148} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1027.** $L_{148} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1028.** $L_{148} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1029.** $L_{148} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1030.** $L_{148} \circ L_{323}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1031.** $L_{148} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1032.** $L_{148} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1033.** $L_{148} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**1034.** $L_{148} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1035.** $L_{148} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1036.** $L_{148} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**1037.** $L_{148} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**1038.** $L_{148} \circ L_{423}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**1039.** $L_{148} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**1040.** $L_{148} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**1041.** $L_{148} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**1042.** $L_{148} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1043.** $L_{148} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1044.** $L_{148} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**1045.** $L_{148} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1046.** $L_{148} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1047.** $L_{148} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1048.** $L_{148} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1049.** $L_{148} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1050.** $L_{150} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1051.** $L_{150} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1052.** $L_{150} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1053.** $L_{150} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1054.** $L_{150} \circ L_{417}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**1055.** $L_{150} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1056.** $L_{150} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1057.** $L_{150} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1058.** $L_{150} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1059.** $L_{152} \circ L_{439}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1060.** $L_{153} \circ L_{156}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1061.** $L_{153} \circ L_{444}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1062.** $L_{156} \circ L_{444}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1063.** $L_{156} \circ L_{450}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1064.** $L_{156} \circ L_{459}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1065.** $L_{156} \circ L_{479}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1066.** $L_{159} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1067.** $L_{159} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1068.** $L_{159} \circ L_{227}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**1069.** $L_{159} \circ L_{254}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1070.** $L_{159} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1071.** $L_{159} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 6 \\ 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 6 & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1072.** $L_{159} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1073.** $L_{159} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1074.** $L_{159} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1075.** $L_{159} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1076.** $L_{159} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**1077.** $L_{159} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6^\nu & 7^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \\ 7^\nu & 6^\nu & 8 & 6 \end{pmatrix}$$

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**1078.** $L_{159} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1079.** $L_{159} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1080.** $L_{159} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**1081.** $L_{159} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 7^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**1082.** $L_{159} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 7^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 6 \\ 6 & 8 & 6^\nu & 7^\nu \\ 8 & 7^\nu & 6 & 6^\nu \end{pmatrix}$$

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**1083.** $L_{159} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**1084.** $L_{159} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1085.** $L_{159} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1086.** $L_{159} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1087.** $L_{159} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1088.** $L_{161} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1089.** $L_{161} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1090.** $L_{161} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1091.** $L_{163} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1092.** $L_{163} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1093.** $L_{163} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1094.** $L_{163} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1095.** $L_{163} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1096.** $L_{163} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1097.** $L_{163} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1098.** $L_{163} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1099.** $L_{163} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1100.** $L_{167} \circ L_{202}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1101.** $L_{167} \circ L_{206}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1102.** $L_{167} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1103.** $L_{167} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \end{pmatrix}$$

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**1104.** $L_{167} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1105.** $L_{167} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1106.** $L_{167} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1107.** $L_{167} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1108.** $L_{167} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**1109.** $L_{167} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1110.** $L_{167} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1111.** $L_{167} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**1112.** $L_{167} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 6 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 6 & 7^\nu & 8 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \end{pmatrix}$$

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**1113.** $L_{167} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1114.** $L_{167} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1115.** $L_{167} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1116.** $L_{167} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 7^\nu & 8 \\ 8 & 6^\nu & 6 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 6 & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1117.** $L_{167} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1118.** $L_{167} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1119.** $L_{167} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1120.** $L_{171} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1121.** $L_{171} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1122.** $L_{171} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1123.** $L_{171} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1124.** $L_{171} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1125.** $L_{171} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1126.** $L_{171} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1127.** $L_{171} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1128.** $L_{171} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1129.** $L_{174} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1130.** $L_{174} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1131.** $L_{174} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**1132.** $L_{176} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 6 \\ 6^\nu & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6^\nu \\ 6 & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**1133.** $L_{176} \circ L_{227}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1134.** $L_{176} \circ L_{232}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1135.** $L_{176} \circ L_{254}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1136.** $L_{176} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1137.** $L_{176} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5^\nu \\ 7 & 8 & 5^\nu & 7 \\ 7 & 5^\nu & 8 & 7 \\ 5^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1138.** $L_{176} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1139.** $L_{176} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1140.** $L_{176} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1141.** $L_{176} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1142.** $L_{176} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**1143.** $L_{176} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 6 \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 6 & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**1144.** $L_{176} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7^\nu & 6 & 8 \end{pmatrix}$$

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**1145.** $L_{176} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 8 & 7 \\ 8 & 7 & 7 & 5^\nu \\ 5^\nu & 7 & 7 & 8 \\ 7 & 8 & 5^\nu & 7 \end{pmatrix}$$

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**1146.** $L_{176} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 8 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 8 & 6 & 6^\nu \end{pmatrix}$$

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**1147.** $L_{176} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1148.** $L_{176} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1149.** $L_{176} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1150.** $L_{176} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1151.** $L_{176} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 8 & 7^\nu \\ 8 & 6 & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 6 & 8 \\ 7^\nu & 8 & 6^\nu & 6 \end{pmatrix}$$

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**1152.** $L_{177} \circ L_{465}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1153.** $L_{179} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1154.** $L_{179} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1155.** $L_{179} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1156.** $L_{179} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1157.** $L_{179} \circ L_{403}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**1158.** $L_{179} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1159.** $L_{179} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1160.** $L_{179} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1161.** $L_{179} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1162.** $L_{181} \circ L_{186}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1163.** $L_{181} \circ L_{193}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**1164.** $L_{181} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1165.** $L_{181} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**1166.** $L_{181} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1167.** $L_{181} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1168.** $L_{181} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1169.** $L_{181} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1170.** $L_{181} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1171.** $L_{181} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1172.** $L_{181} \circ L_{294}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**1173.** $L_{181} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1174.** $L_{181} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**1175.** $L_{181} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**1176.** $L_{181} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1177.** $L_{181} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1178.** $L_{181} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**1179.** $L_{181} \circ L_{401}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1180.** $L_{181} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**1181.** $L_{181} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**1182.** $L_{181} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

---

**1183.** $L_{181} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**1184.** $L_{181} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**1185.** $L_{181} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1186.** $L_{181} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1187.** $L_{181} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1188.** $L_{181} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**1189.** $L_{181} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1190.** $L_{181} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1191.** $L_{181} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1192.** $L_{181} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**1193.** $L_{182} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1194.** $L_{182} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1195.** $L_{182} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**1196.** $L_{182} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1197.** $L_{182} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1198.** $L_{182} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1199.** $L_{182} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1200.** $L_{182} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**1201.** $L_{182} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**1202.** $L_{182} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**1203.** $L_{182} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**1204.** $L_{182} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1205.** $L_{182} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1206.** $L_{182} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**1207.** $L_{182} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1208.** $L_{182} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1209.** $L_{182} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1210.** $L_{183} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1211.** $L_{183} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1212.** $L_{183} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1213.** $L_{183} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1214.** $L_{183} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1215.** $L_{183} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1216.** $L_{183} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**1217.** $L_{183} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**1218.** $L_{183} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \end{pmatrix}$$

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**1219.** $L_{183} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**1220.** $L_{183} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1221.** $L_{183} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1222.** $L_{183} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1223.** $L_{184} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1224.** $L_{184} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1225.** $L_{184} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1226.** $L_{184} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**1227.** $L_{184} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1228.** $L_{184} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1229.** $L_{184} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**1230.** $L_{184} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \end{pmatrix}$$

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**1231.** $L_{184} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \end{pmatrix}$$

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**1232.** $L_{184} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**1233.** $L_{184} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**1234.** $L_{184} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1235.** $L_{184} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1236.** $L_{184} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1237.** $L_{186} \circ L_{206}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1238.** $L_{186} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1239.** $L_{186} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \end{pmatrix}$$

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**1240.** $L_{186} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \end{pmatrix}$$

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**1241.** $L_{186} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1242.** $L_{186} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**1243.** $L_{186} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1244.** $L_{186} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**1245.** $L_{186} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**1246.** $L_{186} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1247.** $L_{186} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \end{pmatrix}$$

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**1248.** $L_{186} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1249.** $L_{186} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**1250.** $L_{186} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1251.** $L_{186} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1252.** $L_{186} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1253.** $L_{186} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 5^\nu & 8 & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1254.** $L_{186} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1255.** $L_{186} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1256.** $L_{186} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1257.** $L_{186} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1258.** $L_{186} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1259.** $L_{190} \circ L_{192}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1260.** $L_{190} \circ L_{442}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1261.** $L_{190} \circ L_{451}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1262.** $L_{190} \circ L_{461}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1263.** $L_{190} \circ L_{478}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1264.** $L_{192} \circ L_{478}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1265.** $L_{193} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1266.** $L_{193} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1267.** $L_{193} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1268.** $L_{193} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1269.** $L_{193} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1270.** $L_{193} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1271.** $L_{193} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1272.** $L_{193} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1273.** $L_{193} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**1274.** $L_{193} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**1275.** $L_{193} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1276.** $L_{193} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1277.** $L_{193} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1278.** $L_{193} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1279.** $L_{193} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1280.** $L_{194} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1281.** $L_{194} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1282.** $L_{194} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1283.** $L_{195} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7^\nu & 5^\nu & 8 \end{pmatrix}$$

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**1284.** $L_{195} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**1285.** $L_{195} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1286.** $L_{195} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1287.** $L_{195} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1288.** $L_{195} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**1289.** $L_{195} \circ L_{323}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1290.** $L_{195} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1291.** $L_{195} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1292.** $L_{195} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1293.** $L_{195} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1294.** $L_{195} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1295.** $L_{195} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1296.** $L_{195} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1297.** $L_{195} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1298.** $L_{195} \circ L_{423}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1299.** $L_{195} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1300.** $L_{195} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1301.** $L_{195} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1302.** $L_{195} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1303.** $L_{195} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1304.** $L_{195} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \\ 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1305.** $L_{195} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1306.** $L_{195} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1307.** $L_{195} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1308.** $L_{195} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**1309.** $L_{195} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**1310.** $L_{196} \circ L_{326}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**1311.** $L_{196} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1312.** $L_{196} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1313.** $L_{196} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1314.** $L_{196} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1315.** $L_{196} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1316.** $L_{196} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1317.** $L_{196} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1318.** $L_{196} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1319.** $L_{197} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \end{pmatrix}$$

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**1320.** $L_{197} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \end{pmatrix}$$

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**1321.** $L_{197} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**1322.** $L_{197} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6^\nu \\ 7^\nu & 6^\nu & 8 & 6 \\ 6^\nu & 7^\nu & 6 & 8 \end{pmatrix}$$

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**1323.** $L_{197} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**1324.** $L_{197} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1325.** $L_{197} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1326.** $L_{197} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1327.** $L_{197} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1328.** $L_{197} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1329.** $L_{197} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1330.** $L_{197} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1331.** $L_{197} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1332.** $L_{197} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \\ 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1333.** $L_{197} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 6 & 8 \\ 7^\nu & 6^\nu & 8 & 6 \\ 8 & 6 & 6^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1334.** $L_{197} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \\ 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1335.** $L_{198} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1336.** $L_{198} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1337.** $L_{198} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1338.** $L_{198} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1339.** $L_{198} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1340.** $L_{198} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1341.** $L_{198} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1342.** $L_{198} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1343.** $L_{198} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1344.** $L_{199} \circ L_{563}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1345.** $L_{202} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1346.** $L_{202} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1347.** $L_{202} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**1348.** $L_{202} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1349.** $L_{202} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1350.** $L_{202} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**1351.** $L_{202} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1352.** $L_{202} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1353.** $L_{202} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1354.** $L_{202} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1355.** $L_{202} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1356.** $L_{202} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1357.** $L_{202} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1358.** $L_{206} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1359.** $L_{206} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1360.** $L_{206} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**1361.** $L_{206} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1362.** $L_{206} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1363.** $L_{206} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1364.** $L_{206} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1365.** $L_{206} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1366.** $L_{206} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1367.** $L_{206} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1368.** $L_{206} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**1369.** $L_{206} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1370.** $L_{213} \circ L_{214}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1371.** $L_{213} \circ L_{555}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1372.** $L_{214} \circ L_{499}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1373.** $L_{214} \circ L_{501}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1374.** $L_{214} \circ L_{553}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1375.** $L_{214} \circ L_{555}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1376.** $L_{216} \circ L_{227}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1377.** $L_{216} \circ L_{232}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1378.** $L_{216} \circ L_{254}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**1379.** $L_{216} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**1380.** $L_{216} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1381.** $L_{216} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1382.** $L_{216} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1383.** $L_{216} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1384.** $L_{216} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 8 & 7^\nu \\ 8 & 6 & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 6 & 8 \\ 7^\nu & 8 & 6^\nu & 6 \end{pmatrix}$$

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**1385.** $L_{216} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**1386.** $L_{216} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 8 & 7^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7^\nu & 8 & 6^\nu & 5^\nu \end{pmatrix}$$

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**1387.** $L_{216} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1388.** $L_{216} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1389.** $L_{216} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 6 & 8 \\ 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 8 & 6 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1390.** $L_{216} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1391.** $L_{216} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1392.** $L_{216} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1393.** $L_{216} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**1394.** $L_{216} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 5^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**1395.** $L_{219} \circ L_{220}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1396.** $L_{219} \circ L_{524}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1397.** $L_{219} \circ L_{526}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1398.** $L_{219} \circ L_{530}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1399.** $L_{219} \circ L_{532}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1400.** $L_{220} \circ L_{530}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1401.** $L_{223} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1402.** $L_{223} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1403.** $L_{223} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**1404.** $L_{223} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1405.** $L_{223} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

---

**1406.** $L_{223} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1407.** $L_{223} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**1408.** $L_{223} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1409.** $L_{223} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1410.** $L_{223} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1411.** $L_{223} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**1412.** $L_{223} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1413.** $L_{223} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1414.** $L_{223} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**1415.** $L_{223} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \end{pmatrix}$$

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**1416.** $L_{223} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \end{pmatrix}$$

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**1417.** $L_{223} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1418.** $L_{223} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1419.** $L_{227} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1420.** $L_{227} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1421.** $L_{227} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1422.** $L_{227} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1423.** $L_{227} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1424.** $L_{227} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1425.** $L_{227} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1426.** $L_{227} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1427.** $L_{227} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1428.** $L_{227} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1429.** $L_{227} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1430.** $L_{232} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**1431.** $L_{232} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1432.** $L_{232} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1433.** $L_{232} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1434.** $L_{232} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1435.** $L_{232} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1436.** $L_{232} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1437.** $L_{232} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1438.** $L_{232} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1439.** $L_{232} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1440.** $L_{232} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1441.** $L_{232} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1442.** $L_{233} \circ L_{540}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1443.** $L_{235} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1444.** $L_{235} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1445.** $L_{235} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1446.** $L_{235} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1447.** $L_{235} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1448.** $L_{235} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1449.** $L_{235} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1450.** $L_{235} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1451.** $L_{235} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1452.** $L_{235} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1453.** $L_{235} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1454.** $L_{235} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1455.** $L_{235} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1456.** $L_{236} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1457.** $L_{236} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**1458.** $L_{236} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1459.** $L_{237} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**1460.** $L_{237} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1461.** $L_{237} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1462.** $L_{237} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1463.** $L_{237} \circ L_{294}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1464.** $L_{237} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1465.** $L_{237} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1466.** $L_{237} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**1467.** $L_{237} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1468.** $L_{237} \circ L_{401}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 8 & 7 & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**1469.** $L_{237} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1470.** $L_{237} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1471.** $L_{237} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1472.** $L_{237} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1473.** $L_{237} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1474.** $L_{237} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1475.** $L_{237} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1476.** $L_{237} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1477.** $L_{237} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1478.** $L_{237} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1479.** $L_{237} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1480.** $L_{237} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1481.** $L_{237} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1482.** $L_{237} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 5^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1483.** $L_{237} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1484.** $L_{238} \circ L_{291}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**1485.** $L_{238} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1486.** $L_{238} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1487.** $L_{238} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1488.** $L_{238} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1489.** $L_{238} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1490.** $L_{238} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1491.** $L_{238} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1492.** $L_{238} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1493.** $L_{239} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1494.** $L_{239} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6^\nu \\ 6 & 8 & 6^\nu & 7^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 7^\nu & 6^\nu & 6 & 8 \end{pmatrix}$$

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**1495.** $L_{239} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \\ 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \end{pmatrix}$$

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**1496.** $L_{239} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \end{pmatrix}$$

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**1497.** $L_{239} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1498.** $L_{239} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1499.** $L_{239} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1500.** $L_{239} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1501.** $L_{239} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1502.** $L_{239} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1503.** $L_{239} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1504.** $L_{239} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1505.** $L_{239} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \\ 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \end{pmatrix}$$

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**1506.** $L_{239} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \\ 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \end{pmatrix}$$

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**1507.** $L_{239} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 6 \\ 7^\nu & 6^\nu & 6 & 8 \\ 6 & 8 & 6^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1508.** $L_{240} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1509.** $L_{240} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1510.** $L_{240} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1511.** $L_{240} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1512.** $L_{240} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1513.** $L_{240} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1514.** $L_{240} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1515.** $L_{240} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1516.** $L_{240} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1517.** $L_{241} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1518.** $L_{241} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1519.** $L_{241} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1520.** $L_{242} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1521.** $L_{242} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1522.** $L_{242} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1523.** $L_{242} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1524.** $L_{242} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1525.** $L_{242} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1526.** $L_{242} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1527.** $L_{242} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1528.** $L_{242} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1529.** $L_{247} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1530.** $L_{247} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1531.** $L_{247} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \end{pmatrix}$$

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**1532.** $L_{247} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1533.** $L_{247} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1534.** $L_{247} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 6 & 6^\nu & 8 \end{pmatrix}$$

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**1535.** $L_{247} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1536.** $L_{247} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1537.** $L_{247} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**1538.** $L_{247} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 6 & 6^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \end{pmatrix}$$

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**1539.** $L_{247} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \end{pmatrix}$$

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**1540.** $L_{247} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1541.** $L_{247} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1542.** $L_{247} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \end{pmatrix}$$

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**1543.** $L_{247} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1544.** $L_{247} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1545.** $L_{254} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1546.** $L_{254} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**1547.** $L_{254} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1548.** $L_{254} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1549.** $L_{254} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1550.** $L_{254} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1551.** $L_{254} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1552.** $L_{254} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1553.** $L_{254} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1554.** $L_{254} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1555.** $L_{254} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**1556.** $L_{256} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1557.** $L_{256} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**1558.** $L_{256} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1559.** $L_{256} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1560.** $L_{256} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1561.** $L_{256} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1562.** $L_{256} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1563.** $L_{256} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1564.** $L_{256} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1565.** $L_{256} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1566.** $L_{256} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1567.** $L_{257} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1568.** $L_{257} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1569.** $L_{257} \circ L_{337}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \end{pmatrix}$$

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**1570.** $L_{257} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1571.** $L_{257} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1572.** $L_{257} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1573.** $L_{257} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1574.** $L_{257} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1575.** $L_{257} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1576.** $L_{259} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1577.** $L_{259} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1578.** $L_{259} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**1579.** $L_{259} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1580.** $L_{259} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1581.** $L_{259} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**1582.** $L_{259} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1583.** $L_{259} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1584.** $L_{259} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**1585.** $L_{259} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1586.** $L_{259} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1587.** $L_{259} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 5^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1588.** $L_{259} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1589.** $L_{260} \circ L_{264}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1590.** $L_{260} \circ L_{520}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1591.** $L_{261} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1592.** $L_{261} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1593.** $L_{261} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1594.** $L_{261} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \end{pmatrix}$$

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**1595.** $L_{261} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \\ 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1596.** $L_{261} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1597.** $L_{261} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1598.** $L_{261} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**1599.** $L_{261} \circ L_{340}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1600.** $L_{261} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1601.** $L_{261} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1602.** $L_{261} \circ L_{382}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**1603.** $L_{261} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1604.** $L_{261} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**1605.** $L_{261} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1606.** $L_{261} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1607.** $L_{261} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1608.** $L_{261} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**1609.** $L_{261} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1610.** $L_{261} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1611.** $L_{261} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1612.** $L_{261} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1613.** $L_{261} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1614.** $L_{261} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \\ 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \end{pmatrix}$$

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**1615.** $L_{261} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1616.** $L_{261} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1617.** $L_{262} \circ L_{518}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1618.** $L_{263} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**1619.** $L_{263} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1620.** $L_{263} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \\ 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \end{pmatrix}$$

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**1621.** $L_{263} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1622.** $L_{263} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**1623.** $L_{263} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**1624.** $L_{263} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1625.** $L_{263} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**1626.** $L_{263} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1627.** $L_{263} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1628.** $L_{263} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1629.** $L_{263} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**1630.** $L_{263} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1631.** $L_{263} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1632.** $L_{263} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1633.** $L_{263} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1634.** $L_{263} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1635.** $L_{263} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \end{pmatrix}$$

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**1636.** $L_{264} \circ L_{511}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1637.** $L_{264} \circ L_{520}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1638.** $L_{264} \circ L_{541}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1639.** $L_{264} \circ L_{552}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1640.** $L_{265} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1641.** $L_{265} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1642.** $L_{265} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6^\nu \\ 6 & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 6 \\ 6^\nu & 6 & 7^\nu & 8 \end{pmatrix}$$

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**1643.** $L_{265} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1644.** $L_{265} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1645.** $L_{265} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6^\nu \\ 6 & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**1646.** $L_{265} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**1647.** $L_{265} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6^\nu & 7^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \\ 7^\nu & 6^\nu & 8 & 6 \end{pmatrix}$$

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**1648.** $L_{265} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 6 & 7^\nu \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 7^\nu & 6 & 8 & 6^\nu \end{pmatrix}$$

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**1649.** $L_{265} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5^\nu & 7 \\ 5^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 5^\nu \\ 7 & 5^\nu & 8 & 7 \end{pmatrix}$$

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**1650.** $L_{265} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1651.** $L_{265} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1652.** $L_{265} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1653.** $L_{265} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1654.** $L_{267} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1655.** $L_{267} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1656.** $L_{267} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1657.** $L_{267} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1658.** $L_{267} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1659.** $L_{267} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1660.** $L_{267} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1661.** $L_{267} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1662.** $L_{267} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1663.** $L_{268} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1664.** $L_{268} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**1665.** $L_{268} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1666.** $L_{273} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 8 & 6 & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**1667.** $L_{273} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1668.** $L_{273} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1669.** $L_{273} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1670.** $L_{273} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1671.** $L_{273} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1672.** $L_{273} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1673.** $L_{273} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1674.** $L_{273} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1675.** $L_{273} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1676.** $L_{273} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**1677.** $L_{273} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1678.** $L_{274} \circ L_{277}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1679.** $L_{274} \circ L_{493}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1680.** $L_{275} \circ L_{495}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1681.** $L_{276} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1682.** $L_{276} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1683.** $L_{276} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1684.** $L_{276} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \\ 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1685.** $L_{276} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1686.** $L_{276} \circ L_{340}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**1687.** $L_{276} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1688.** $L_{276} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**1689.** $L_{276} \circ L_{382}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1690.** $L_{276} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1691.** $L_{276} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1692.** $L_{276} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**1693.** $L_{276} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1694.** $L_{276} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**1695.** $L_{276} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1696.** $L_{276} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1697.** $L_{276} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1698.** $L_{276} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1699.** $L_{276} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1700.** $L_{276} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1701.** $L_{276} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1702.** $L_{276} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1703.** $L_{276} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 8 & 6 & 7^\nu & 6 \\ 6 & 7^\nu & 6 & 8 \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1704.** $L_{277} \circ L_{488}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1705.** $L_{277} \circ L_{493}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1706.** $L_{277} \circ L_{568}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1707.** $L_{277} \circ L_{569}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1708.** $L_{278} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \\ 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \end{pmatrix}$$

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**1709.** $L_{278} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**1710.** $L_{278} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{array}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1711.** $L_{278} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1712.** $L_{278} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**1713.** $L_{278} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1714.** $L_{278} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**1715.** $L_{278} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1716.** $L_{278} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**1717.** $L_{278} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1718.** $L_{278} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1719.** $L_{278} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1720.** $L_{278} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1721.** $L_{278} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1722.** $L_{278} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1723.** $L_{278} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1724.** $L_{279} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1725.** $L_{279} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1726.** $L_{279} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1727.** $L_{279} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1728.** $L_{279} \circ L_{379}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7 & 7 & 8 \\ 7 & 5^\nu & 8 & 7 \\ 7 & 8 & 5^\nu & 7 \\ 8 & 7 & 7 & 5^\nu \end{pmatrix}$$

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**1729.** $L_{279} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1730.** $L_{279} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1731.** $L_{279} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1732.** $L_{279} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1733.** $L_{282} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1734.** $L_{282} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1735.** $L_{282} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1736.** $L_{282} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1737.** $L_{282} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1738.** $L_{282} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1739.** $L_{282} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1740.** $L_{282} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1741.** $L_{282} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1742.** $L_{282} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \end{pmatrix}$$

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**1743.** $L_{286} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \end{pmatrix}$$

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**1744.** $L_{286} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1745.** $L_{286} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7^\nu & 6 & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1746.** $L_{286} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1747.** $L_{286} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1748.** $L_{286} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**1749.** $L_{286} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**1750.** $L_{286} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1751.** $L_{286} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1752.** $L_{286} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \end{pmatrix}$$

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**1753.** $L_{289} \circ L_{342}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1754.** $L_{289} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1755.** $L_{289} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1756.** $L_{289} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1757.** $L_{289} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1758.** $L_{289} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1759.** $L_{289} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1760.** $L_{289} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1761.** $L_{289} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1762.** $L_{290} \circ L_{327}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \end{pmatrix}$$

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**1763.** $L_{290} \circ L_{341}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 6 & 7^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \end{pmatrix}$$

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**1764.** $L_{290} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1765.** $L_{290} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1766.** $L_{290} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1767.** $L_{290} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1768.** $L_{290} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1769.** $L_{290} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1770.** $L_{290} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1771.** $L_{290} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1772.** $L_{290} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1773.** $L_{290} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5^\nu & 8 \\ 7 & 7 & 8 & 5^\nu \\ 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1774.** $L_{290} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 6 & 8 \\ 6^\nu & 7^\nu & 8 & 6 \\ 8 & 6 & 7^\nu & 6^\nu \\ 6 & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1775.** $L_{291} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1776.** $L_{291} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1777.** $L_{291} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1778.** $L_{291} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1779.** $L_{291} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1780.** $L_{291} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1781.** $L_{291} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1782.** $L_{291} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1783.** $L_{292} \circ L_{303}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 6^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6^\nu & 5^\nu & 8 \end{pmatrix}$$

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**1784.** $L_{292} \circ L_{323}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**1785.** $L_{292} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**1786.** $L_{292} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**1787.** $L_{292} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1788.** $L_{292} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1789.** $L_{292} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1790.** $L_{292} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1791.** $L_{292} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1792.** $L_{292} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1793.** $L_{292} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1794.** $L_{292} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1795.** $L_{292} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1796.** $L_{292} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1797.** $L_{292} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1798.** $L_{292} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1799.** $L_{292} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1800.** $L_{292} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1801.** $L_{292} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1802.** $L_{292} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1803.** $L_{292} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1804.** $L_{292} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1805.** $L_{292} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1806.** $L_{292} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1807.** $L_{292} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1808.** $L_{292} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1809.** $L_{292} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1810.** $L_{292} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1811.** $L_{292} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**1812.** $L_{292} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**1813.** $L_{292} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**1814.** $L_{292} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**1815.** $L_{292} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1816.** $L_{294} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1817.** $L_{294} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**1818.** $L_{294} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**1819.** $L_{294} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**1820.** $L_{294} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1821.** $L_{294} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1822.** $L_{296} \circ L_{538}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**1823.** $L_{300} \circ L_{547}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**1824.** $L_{303} \circ L_{315}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1825.** $L_{303} \circ L_{320}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1826.** $L_{303} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**1827.** $L_{303} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**1828.** $L_{303} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1829.** $L_{303} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1830.** $L_{303} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1831.** $L_{303} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1832.** $L_{303} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1833.** $L_{303} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1834.** $L_{303} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1835.** $L_{303} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1836.** $L_{303} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**1837.** $L_{303} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1838.** $L_{303} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1839.** $L_{303} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1840.** $L_{303} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1841.** $L_{303} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1842.** $L_{303} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1843.** $L_{303} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1844.** $L_{303} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1845.** $L_{303} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1846.** $L_{303} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1847.** $L_{303} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**1848.** $L_{303} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**1849.** $L_{303} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**1850.** $L_{303} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**1851.** $L_{303} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1852.** $L_{304} \circ L_{342}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1853.** $L_{304} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**1854.** $L_{304} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1855.** $L_{307} \circ L_{309}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1856.** $L_{307} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1857.** $L_{307} \circ L_{348}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1858.** $L_{307} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1859.** $L_{307} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1860.** $L_{307} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1861.** $L_{307} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1862.** $L_{307} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**1863.** $L_{307} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**1864.** $L_{309} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1865.** $L_{309} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1866.** $L_{311} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1867.** $L_{311} \circ L_{346}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \end{pmatrix}$$

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**1868.** $L_{311} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**1869.** $L_{311} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \end{pmatrix}$$

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**1870.** $L_{311} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 5^\nu & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**1871.** $L_{311} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**1872.** $L_{311} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1873.** $L_{313} \circ L_{315}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1874.** $L_{313} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1875.** $L_{313} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1876.** $L_{315} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**1877.** $L_{315} \circ L_{369}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1878.** $L_{315} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1879.** $L_{315} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

---

**1880.** $L_{315} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1881.** $L_{315} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1882.** $L_{315} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1883.** $L_{320} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 6 & 7^\nu & 8 \\ 7^\nu & 6^\nu & 8 & 6 \\ 6 & 8 & 6^\nu & 7^\nu \\ 8 & 7^\nu & 6 & 6^\nu \end{pmatrix}$$

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**1884.** $L_{320} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1885.** $L_{320} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1886.** $L_{320} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 6 \\ 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 6 & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1887.** $L_{320} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1888.** $L_{320} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1889.** $L_{321} \circ L_{561}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1890.** $L_{323} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1891.** $L_{323} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1892.** $L_{323} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1893.** $L_{323} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1894.** $L_{323} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1895.** $L_{325} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**1896.** $L_{325} \circ L_{337}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1897.** $L_{325} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1898.** $L_{325} \circ L_{346}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1899.** $L_{325} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1900.** $L_{325} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \end{pmatrix}$$

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**1901.** $L_{325} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**1902.** $L_{325} \circ L_{401}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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**1903.** $L_{325} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1904.** $L_{325} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1905.** $L_{325} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1906.** $L_{325} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1907.** $L_{325} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1908.** $L_{325} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1909.** $L_{325} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1910.** $L_{325} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1911.** $L_{325} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1912.** $L_{325} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1913.** $L_{325} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1914.** $L_{325} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1915.** $L_{325} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1916.** $L_{325} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1917.** $L_{325} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**1918.** $L_{325} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1919.** $L_{325} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1920.** $L_{325} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1921.** $L_{325} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1922.** $L_{325} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1923.** $L_{325} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1924.** $L_{325} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1925.** $L_{325} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1926.** $L_{325} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1927.** $L_{326} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1928.** $L_{326} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1929.** $L_{326} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1930.** $L_{326} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1931.** $L_{326} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1932.** $L_{326} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1933.** $L_{326} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1934.** $L_{326} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1935.** $L_{327} \circ L_{383}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 7^\nu & 6 & 8 & 6^\nu \end{pmatrix}$$

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**1936.** $L_{327} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1937.** $L_{327} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1938.** $L_{327} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1939.** $L_{327} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1940.** $L_{327} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1941.** $L_{327} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1942.** $L_{327} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1943.** $L_{327} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**1944.** $L_{327} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \end{pmatrix}$$

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**1945.** $L_{327} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 6 \\ 6^\nu & 7^\nu & 6 & 8 \\ 6 & 8 & 7^\nu & 6^\nu \\ 8 & 6 & 6^\nu & 7^\nu \end{pmatrix}$$

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**1946.** $L_{327} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \\ 5^\nu & 8 & 7 & 7 \\ 8 & 5^\nu & 7 & 7 \end{pmatrix}$$

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**1947.** $L_{328} \circ L_{384}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1948.** $L_{328} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1949.** $L_{328} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1950.** $L_{328} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1951.** $L_{328} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1952.** $L_{328} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1953.** $L_{328} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1954.** $L_{328} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1955.** $L_{328} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1956.** $L_{329} \circ L_{384}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

---

**1957.** $L_{329} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**1958.** $L_{329} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1959.** $L_{330} \circ L_{337}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1960.** $L_{330} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1961.** $L_{330} \circ L_{346}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1962.** $L_{330} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1963.** $L_{330} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**1964.** $L_{330} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**1965.** $L_{330} \circ L_{386}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1966.** $L_{330} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**1967.** $L_{330} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1968.** $L_{330} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1969.** $L_{330} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

---

**1970.** $L_{330} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1971.** $L_{330} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1972.** $L_{330} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1973.** $L_{330} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**1974.** $L_{330} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**1975.** $L_{330} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1976.** $L_{330} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1977.** $L_{330} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**1978.** $L_{330} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**1979.** $L_{330} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1980.** $L_{330} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**1981.** $L_{330} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**1982.** $L_{330} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**1983.** $L_{330} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1984.** $L_{330} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 5^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**1985.** $L_{334} \circ L_{574}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**1986.** $L_{337} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1987.** $L_{337} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**1988.** $L_{337} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1989.** $L_{337} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1990.** $L_{337} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1991.** $L_{337} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**1992.** $L_{339} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 5^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 5^\nu & 6^\nu & 8 \end{pmatrix}$$

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**1993.** $L_{339} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1994.** $L_{339} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1995.** $L_{339} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1996.** $L_{339} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1997.** $L_{339} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**1998.** $L_{339} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**1999.** $L_{339} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2000.** $L_{339} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**2001.** $L_{339} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

---

**2002.** $L_{339} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**2003.** $L_{339} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**2004.** $L_{339} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2005.** $L_{339} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2006.** $L_{339} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2007.** $L_{339} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2008.** $L_{339} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2009.** $L_{339} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2010.** $L_{339} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2011.** $L_{339} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2012.** $L_{339} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2013.** $L_{339} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**2014.** $L_{339} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**2015.** $L_{339} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**2016.** $L_{339} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**2017.** $L_{339} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2018.** $L_{339} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2019.** $L_{339} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2020.** $L_{340} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**2021.** $L_{340} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2022.** $L_{340} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2023.** $L_{340} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**2024.** $L_{340} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**2025.** $L_{340} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2026.** $L_{341} \circ L_{383}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2027.** $L_{341} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2028.** $L_{341} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2029.** $L_{341} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**2030.** $L_{341} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2031.** $L_{341} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \end{pmatrix}$$

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**2032.** $L_{342} \circ L_{359}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2033.** $L_{342} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2034.** $L_{342} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**2035.** $L_{342} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**2036.** $L_{342} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2037.** $L_{342} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2038.** $L_{342} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**2039.** $L_{343} \circ L_{515}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**2040.** $L_{346} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2041.** $L_{346} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2042.** $L_{346} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \end{pmatrix}$$

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**2043.** $L_{346} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6^\nu & 8 \\ 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 8 & 7^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2044.** $L_{346} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 6 & 8 \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2045.** $L_{346} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2046.** $L_{346} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2047.** $L_{346} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \end{pmatrix}$$

---

**2048.** $L_{346} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2049.** $L_{346} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2050.** $L_{348} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2051.** $L_{348} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

**2052.** $L_{350} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

---

**2053.** $L_{350} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

---

**2054.** $L_{350} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2055.** $L_{350} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2056.** $L_{350} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2057.** $L_{350} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2058.** $L_{358} \circ L_{507}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**2059.** $L_{359} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2060.** $L_{359} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2061.** $L_{360} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2062.** $L_{360} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2063.** $L_{360} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2064.** $L_{360} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2065.** $L_{360} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2066.** $L_{360} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2067.** $L_{360} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2068.** $L_{360} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2069.** $L_{360} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2070.** $L_{360} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2071.** $L_{360} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**2072.** $L_{360} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2073.** $L_{360} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2074.** $L_{360} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2075.** $L_{360} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2076.** $L_{360} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2077.** $L_{360} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2078.** $L_{360} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2079.** $L_{360} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2080.** $L_{360} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**2081.** $L_{360} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 6^\nu & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**2082.** $L_{360} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**2083.** $L_{360} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 6 & 8 \\ 6 & 7^\nu & 8 & 6 \\ 6 & 8 & 7^\nu & 6 \\ 8 & 6 & 6 & 7^\nu \end{pmatrix}$$

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**2084.** $L_{360} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2085.** $L_{360} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2086.** $L_{363} \circ L_{482}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**2087.** $L_{367} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**2088.** $L_{367} \circ L_{386}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**2089.** $L_{367} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2090.** $L_{367} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2091.** $L_{367} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2092.** $L_{367} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2093.** $L_{367} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2094.** $L_{367} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2095.** $L_{367} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2096.** $L_{367} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2097.** $L_{367} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2098.** $L_{367} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2099.** $L_{367} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**2100.** $L_{367} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**2101.** $L_{367} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2102.** $L_{367} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2103.** $L_{367} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2104.** $L_{367} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2105.** $L_{367} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2106.** $L_{367} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**2107.** $L_{367} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**2108.** $L_{367} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**2109.** $L_{368} \circ L_{384}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2110.** $L_{368} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

---

**2111.** $L_{368} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**2112.** $L_{369} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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**2113.** $L_{369} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2114.** $L_{371} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2115.** $L_{371} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2116.** $L_{371} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2117.** $L_{371} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2118.** $L_{376} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5^\nu \\ 7 & 8 & 5^\nu & 7 \\ 7 & 5^\nu & 8 & 7 \\ 5^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2119.** $L_{376} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6^\nu & 8 & 6 \\ 8 & 7^\nu & 6 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 6 & 8 & 6^\nu & 7^\nu \end{pmatrix}$$

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**2120.** $L_{376} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6^\nu \\ 8 & 7^\nu & 6^\nu & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 6^\nu & 8 & 6 & 7^\nu \end{pmatrix}$$

---

**2121.** $L_{376} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5^\nu & 8 & 7 \\ 8 & 7 & 7 & 5^\nu \\ 5^\nu & 7 & 7 & 8 \\ 7 & 8 & 5^\nu & 7 \end{pmatrix}$$

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**2122.** $L_{376} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2123.** $L_{376} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2124.** $L_{376} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2125.** $L_{376} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2126.** $L_{377} \circ L_{492}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**2127.** $L_{379} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**2128.** $L_{379} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**2129.** $L_{379} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**2130.** $L_{379} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2131.** $L_{381} \circ L_{401}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2132.** $L_{381} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2133.** $L_{381} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2134.** $L_{381} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

---

**2135.** $L_{381} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

---

**2136.** $L_{381} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

---

**2137.** $L_{381} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2138.** $L_{381} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2139.** $L_{381} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2140.** $L_{381} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

---

**2141.** $L_{381} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**2142.** $L_{381} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**2143.** $L_{381} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2144.** $L_{381} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2145.** $L_{381} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2146.** $L_{381} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2147.** $L_{381} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2148.** $L_{381} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**2149.** $L_{381} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2150.** $L_{381} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2151.** $L_{381} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \end{pmatrix}$$

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**2152.** $L_{381} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6 & 7^\nu & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \end{pmatrix}$$

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**2153.** $L_{382} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2154.** $L_{382} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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**2155.** $L_{382} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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**2156.** $L_{382} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2157.** $L_{382} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2158.** $L_{382} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \end{pmatrix}$$

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**2159.** $L_{383} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2160.** $L_{383} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2161.** $L_{383} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2162.** $L_{383} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 8 & 6^\nu & 5^\nu & 7^\nu \end{pmatrix}$$

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**2163.** $L_{383} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7^\nu & 6 \\ 8 & 6^\nu & 6 & 7^\nu \end{pmatrix}$$

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**2164.** $L_{384} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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**2165.** $L_{384} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2166.** $L_{384} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2167.** $L_{384} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**2168.** $L_{384} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 8 & 6^\nu & 7^\nu & 5^\nu \end{pmatrix}$$

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**2169.** $L_{384} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \\ 6^\nu & 8 & 7 & 7 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2170.** $L_{386} \circ L_{393}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2171.** $L_{386} \circ L_{416}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2172.** $L_{386} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \end{pmatrix}$$

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**2173.** $L_{386} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7^\nu & 6^\nu & 5^\nu \end{pmatrix}$$

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**2174.** $L_{386} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**2175.** $L_{386} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2176.** $L_{386} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2177.** $L_{390} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2178.** $L_{390} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2179.** $L_{390} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**2180.** $L_{391} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \end{pmatrix}$$

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**2181.** $L_{391} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \end{pmatrix}$$

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**2182.** $L_{391} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2183.** $L_{391} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \end{pmatrix}$$

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**2184.** $L_{391} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 8 & 5^\nu & 6^\nu & 7^\nu \end{pmatrix}$$

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**2185.** $L_{393} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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**2186.** $L_{393} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2187.** $L_{398} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2188.** $L_{398} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2189.** $L_{398} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2190.** $L_{398} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2191.** $L_{398} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2192.** $L_{400} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7^\nu & 6 & 8 \\ 6 & 6^\nu & 8 & 7^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 8 & 6 & 7^\nu & 6^\nu \end{pmatrix}$$

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**2193.** $L_{400} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5^\nu & 7 \\ 5^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 5^\nu \\ 7 & 5^\nu & 8 & 7 \end{pmatrix}$$

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**2194.** $L_{400} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \\ 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2195.** $L_{400} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2196.** $L_{400} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2197.** $L_{400} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2198.** $L_{400} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2199.** $L_{400} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 6 \\ 6^\nu & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6^\nu \\ 6 & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2200.** $L_{401} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2201.** $L_{401} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2202.** $L_{401} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7^\nu & 8 & 5^\nu \end{pmatrix}$$

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**2203.** $L_{401} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6 & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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**2204.** $L_{403} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2205.** $L_{403} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2206.** $L_{403} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2207.** $L_{403} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2208.** $L_{405} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2209.** $L_{405} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2210.** $L_{405} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**2211.** $L_{405} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2212.** $L_{405} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2213.** $L_{405} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2214.** $L_{405} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2215.** $L_{405} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2216.** $L_{405} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2217.** $L_{405} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2218.** $L_{405} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**2219.** $L_{405} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2220.** $L_{405} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2221.** $L_{405} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2222.** $L_{405} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2223.** $L_{405} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2224.** $L_{405} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2225.** $L_{405} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2226.** $L_{405} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2227.** $L_{405} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2228.** $L_{406} \circ L_{470}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2229.** $L_{407} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \end{pmatrix}$$

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**2230.** $L_{407} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2231.** $L_{407} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2232.** $L_{407} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2233.** $L_{407} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2234.** $L_{407} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 5^\nu & 6^\nu \\ 8 & 7^\nu & 6^\nu & 5^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2235.** $L_{407} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2236.** $L_{407} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**2237.** $L_{407} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2238.** $L_{407} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2239.** $L_{407} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2240.** $L_{407} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2241.** $L_{407} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2242.** $L_{407} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2243.** $L_{407} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2244.** $L_{407} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2245.** $L_{407} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2246.** $L_{408} \circ L_{472}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2247.** $L_{409} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2248.** $L_{409} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 6 \\ 6^\nu & 7^\nu & 6 & 8 \\ 8 & 6 & 7^\nu & 6^\nu \\ 6 & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2249.** $L_{409} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2250.** $L_{409} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2251.** $L_{411} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2252.** $L_{411} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2253.** $L_{411} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2254.** $L_{411} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**2255.** $L_{411} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2256.** $L_{416} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2257.** $L_{416} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2258.** $L_{417} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2259.** $L_{417} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2260.** $L_{417} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2261.** $L_{417} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \end{pmatrix}$$

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**2262.** $L_{419} \circ L_{433}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2263.** $L_{420} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2264.** $L_{420} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2265.** $L_{420} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2266.** $L_{420} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2267.** $L_{420} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6^\nu & 5^\nu & 7^\nu & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2268.** $L_{420} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2269.** $L_{420} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**2270.** $L_{420} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2271.** $L_{420} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2272.** $L_{420} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2273.** $L_{420} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2274.** $L_{420} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2275.** $L_{420} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2276.** $L_{420} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2277.** $L_{420} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2278.** $L_{420} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2279.** $L_{420} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2280.** $L_{420} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2281.** $L_{421} \circ L_{435}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2282.** $L_{422} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2283.** $L_{422} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7^\nu & 5^\nu & 6^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2284.** $L_{422} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 5^\nu & 6^\nu & 7^\nu & 8 \\ 6^\nu & 5^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2285.** $L_{422} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6 \\ 8 & 7^\nu & 6 & 6 \\ 6 & 6 & 7^\nu & 8 \\ 6 & 6 & 8 & 7^\nu \end{pmatrix}$$

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**2286.** $L_{422} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2287.** $L_{422} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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**2288.** $L_{422} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2289.** $L_{422} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2290.** $L_{422} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2291.** $L_{422} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2292.** $L_{422} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 7 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2293.** $L_{422} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2294.** $L_{422} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2295.** $L_{422} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{array}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2296.** $L_{422} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{array}{c} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{array}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 7 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2297.** $L_{423} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 5^\nu \\ 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 5^\nu & 6^\nu & 8 & 7^\nu \end{pmatrix}$$

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**2298.** $L_{423} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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**2299.** $L_{423} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2300.** $L_{423} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 8 & 5^\nu & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 5^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2301.** $L_{426} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \\ 8 & 7 & 5^\nu & 7 \end{pmatrix}$$

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**2302.** $L_{426} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2303.** $L_{426} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2304.** $L_{426} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6^\nu & 6 \\ 6 & 7^\nu & 8 & 6^\nu \\ 6^\nu & 6 & 7^\nu & 8 \\ 8 & 6^\nu & 6 & 7^\nu \end{pmatrix}$$

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**2305.** $L_{426} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \\ 8 & 6 & 6^\nu & 7^\nu \end{pmatrix}$$

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**2306.** $L_{426} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2307.** $L_{426} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 8 & 6^\nu & 7 & 7 \end{pmatrix}$$

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**2308.** $L_{430} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2309.** $L_{430} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2310.** $L_{430} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2311.** $L_{430} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 8 & 7 & 7 & 6^\nu \end{pmatrix}$$

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**2312.** $L_{431} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2313.** $L_{431} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6^\nu & 7 & 8 \\ 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2314.** $L_{434} \circ L_{436}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 5^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2315.** $L_{434} \circ L_{447}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 5^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 5^\nu & 7^\nu & 8 \end{pmatrix}$$

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**2316.** $L_{434} \circ L_{469}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2317.** $L_{434} \circ L_{471}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 5^\nu & 7^\nu & 8 \end{pmatrix}$$

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**2318.** $L_{434} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2319.** $L_{434} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**2320.** $L_{434} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2321.** $L_{434} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**2322.** $L_{434} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2323.** $L_{434} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2324.** $L_{434} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2325.** $L_{434} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2326.** $L_{434} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2327.** $L_{434} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2328.** $L_{434} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7^\nu & 8 & 6^\nu & 5^\nu \\ 5^\nu & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2329.** $L_{434} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2330.** $L_{434} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2331.** $L_{434} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2332.** $L_{434} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2333.** $L_{434} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2334.** $L_{436} \circ L_{469}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 5^\nu & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**2335.** $L_{436} \circ L_{471}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6 \\ 7^\nu & 8 & 6 & 6 \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2336.** $L_{436} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2337.** $L_{436} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2338.** $L_{436} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**2339.** $L_{436} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2340.** $L_{436} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2341.** $L_{436} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2342.** $L_{436} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2343.** $L_{436} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2344.** $L_{436} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2345.** $L_{436} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2346.** $L_{436} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2347.** $L_{436} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2348.** $L_{436} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2349.** $L_{436} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2350.** $L_{436} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**2351.** $L_{437} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2352.** $L_{437} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2353.** $L_{437} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2354.** $L_{437} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2355.** $L_{442} \circ L_{478}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2356.** $L_{444} \circ L_{450}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2357.** $L_{444} \circ L_{459}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2358.** $L_{444} \circ L_{479}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2359.** $L_{447} \circ L_{464}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6 & 6^\nu \\ 6 & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 6 \\ 6^\nu & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2360.** $L_{447} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**2361.** $L_{447} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2362.** $L_{447} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2363.** $L_{447} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2364.** $L_{447} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2365.** $L_{451} \circ L_{478}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2366.** $L_{451} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2367.** $L_{451} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2368.** $L_{451} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2369.** $L_{451} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2370.** $L_{455} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 6 \\ 6 & 8 & 7^\nu & 6^\nu \\ 6^\nu & 6 & 8 & 7^\nu \\ 7^\nu & 6^\nu & 6 & 8 \end{pmatrix}$$

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**2371.** $L_{455} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2372.** $L_{455} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2373.** $L_{455} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5^\nu & 7 \\ 7 & 8 & 7 & 5^\nu \\ 5^\nu & 7 & 8 & 7 \\ 7 & 5^\nu & 7 & 8 \end{pmatrix}$$

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**2374.** $L_{455} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2375.** $L_{455} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2376.** $L_{459} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2377.** $L_{459} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2378.** $L_{459} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2379.** $L_{459} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2380.** $L_{461} \circ L_{478}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2381.** $L_{464} \circ L_{503}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5^\nu \\ 7 & 8 & 5^\nu & 7 \\ 7 & 5^\nu & 8 & 7 \\ 5^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2382.** $L_{464} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2383.** $L_{464} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2384.** $L_{464} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2385.** $L_{464} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2386.** $L_{467} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2387.** $L_{467} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2388.** $L_{467} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2389.** $L_{467} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2390.** $L_{469} \circ L_{471}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 5^\nu \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6 & 6 & 8 & 7^\nu \\ 6 & 6 & 7^\nu & 8 \end{pmatrix}$$

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**2391.** $L_{469} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7^\nu & 6^\nu & 5^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 5^\nu & 8 & 7^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2392.** $L_{469} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6 & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2393.** $L_{469} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2394.** $L_{469} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2395.** $L_{469} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2396.** $L_{469} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**2397.** $L_{469} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7^\nu & 8 & 5^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 5^\nu & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**2398.** $L_{469} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2399.** $L_{469} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2400.** $L_{469} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2401.** $L_{469} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2402.** $L_{469} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2403.** $L_{469} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2404.** $L_{469} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2405.** $L_{469} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2406.** $L_{469} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2407.** $L_{469} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**2408.** $L_{471} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2409.** $L_{471} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2410.** $L_{471} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**2411.** $L_{471} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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**2412.** $L_{471} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2413.** $L_{471} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2414.** $L_{471} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2415.** $L_{471} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2416.** $L_{471} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2417.** $L_{471} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6 & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2418.** $L_{471} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2419.** $L_{471} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 7 & 8 \\ 6^\nu & 7 & 8 & 7 \end{pmatrix}$$

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**2420.** $L_{471} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2421.** $L_{471} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 6^\nu & 8 & 7 \end{pmatrix}$$

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**2422.** $L_{471} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2423.** $L_{474} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2424.** $L_{474} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2425.** $L_{474} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7^\nu & 6^\nu & 5^\nu & 8 \end{pmatrix}$$

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**2426.** $L_{474} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6 & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**2427.** $L_{474} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 6^\nu & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2428.** $L_{487} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 5^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**2429.** $L_{487} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2430.** $L_{487} \circ L_{513}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2431.** $L_{487} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2432.** $L_{487} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2433.** $L_{487} \circ L_{526}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2434.** $L_{487} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2435.** $L_{487} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**2436.** $L_{487} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**2437.** $L_{487} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 5^\nu & 8 & 6^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6^\nu & 7^\nu & 5^\nu & 8 \end{pmatrix}$$

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**2438.** $L_{487} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**2439.** $L_{488} \circ L_{493}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2440.** $L_{490} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2441.** $L_{490} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2442.** $L_{490} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2443.** $L_{490} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 8 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2444.** $L_{493} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2445.** $L_{493} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2446.** $L_{494} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 5^\nu \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2447.** $L_{494} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 5^\nu & 6^\nu & 8 \end{pmatrix}$$

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**2448.** $L_{494} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2449.** $L_{494} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7^\nu & 6 \\ 6^\nu & 8 & 6 & 7^\nu \\ 6 & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6^\nu & 8 \end{pmatrix}$$

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**2450.** $L_{496} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2451.** $L_{496} \circ L_{513}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2452.** $L_{496} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2453.** $L_{496} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7^\nu & 5^\nu & 8 & 6^\nu \\ 5^\nu & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**2454.** $L_{496} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2455.** $L_{496} \circ L_{526}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2456.** $L_{496} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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**2457.** $L_{496} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2458.** $L_{496} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2459.** $L_{496} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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**2460.** $L_{496} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6 \\ 6^\nu & 8 & 5^\nu & 7^\nu \\ 7^\nu & 6 & 8 & 6 \\ 5^\nu & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**2461.** $L_{499} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2462.** $L_{499} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2463.** $L_{499} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2464.** $L_{499} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2465.** $L_{499} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2466.** $L_{501} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2467.** $L_{503} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2468.** $L_{503} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2469.** $L_{503} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2470.** $L_{503} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2471.** $L_{503} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7^\nu & 6^\nu \\ 7^\nu & 8 & 6^\nu & 6 \\ 6 & 6^\nu & 8 & 7^\nu \\ 6^\nu & 7^\nu & 6 & 8 \end{pmatrix}$$

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**2472.** $L_{511} \circ L_{520}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2473.** $L_{512} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 5^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 5^\nu & 6^\nu & 8 \end{pmatrix}$$

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**2474.** $L_{512} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 5^\nu & 8 & 7^\nu & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**2475.** $L_{512} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**2476.** $L_{512} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2477.** $L_{512} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2478.** $L_{512} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2479.** $L_{513} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2480.** $L_{513} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 7 & 7 & 8 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \end{pmatrix}$$

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**2481.** $L_{517} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 5^\nu & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2482.** $L_{517} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**2483.** $L_{517} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 5^\nu \\ 5^\nu & 7^\nu & 8 & 6^\nu \\ 7^\nu & 6 & 6 & 8 \end{pmatrix}$$

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**2484.** $L_{517} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2485.** $L_{517} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2486.** $L_{517} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2487.** $L_{519} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 6 & 7^\nu \\ 6^\nu & 8 & 7^\nu & 6 \\ 7^\nu & 6 & 8 & 6^\nu \\ 6 & 7^\nu & 6^\nu & 8 \end{pmatrix}$$

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**2488.** $L_{520} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2489.** $L_{520} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2490.** $L_{522} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6^\nu & 7^\nu \\ 7^\nu & 8 & 6 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 6 \\ 6 & 6^\nu & 7^\nu & 8 \end{pmatrix}$$

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**2491.** $L_{522} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2492.** $L_{522} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2493.** $L_{524} \circ L_{530}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2494.** $L_{526} \circ L_{530}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2495.** $L_{526} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2496.** $L_{526} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6^\nu & 7 & 7 \\ 7 & 7 & 6^\nu & 8 \\ 7 & 7 & 8 & 6^\nu \\ 6^\nu & 8 & 7 & 7 \end{pmatrix}$$

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**2497.** $L_{530} \circ L_{532}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6^\nu & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2498.** $L_{535} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 6^\nu & 7^\nu \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7^\nu & 8 & 5^\nu \\ 7 & 6^\nu & 7 & 8 \end{pmatrix}$$

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**2499.** $L_{535} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6^\nu & 7 \\ 7 & 7 & 8 & 6 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2500.** $L_{542} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7 & 7 \\ 5^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 5^\nu \\ 7 & 7 & 5^\nu & 8 \end{pmatrix}$$

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**2501.** $L_{544} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 6^\nu & 7^\nu \\ 6 & 8 & 7^\nu & 6 \\ 6 & 7^\nu & 8 & 6 \\ 7^\nu & 6^\nu & 5^\nu & 8 \end{pmatrix}$$

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**2502.** $L_{544} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6^\nu \\ 6^\nu & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**2503.** $L_{544} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2504.** $L_{551} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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**2505.** $L_{551} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6^\nu & 8 & 7 & 7 \\ 7 & 7 & 8 & 6^\nu \\ 7 & 7 & 6^\nu & 8 \end{pmatrix}$$

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**2506.** $L_{553} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7 & 7 & 8 \end{pmatrix}$$

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**2507.** $L_{560} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7^\nu & 6^\nu \\ 7 & 8 & 6^\nu & 7 \\ 7 & 6^\nu & 8 & 7 \\ 6^\nu & 7^\nu & 5^\nu & 8 \end{pmatrix}$$

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**2508.** $L_{565} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \qquad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5^\nu & 7^\nu & 6^\nu \\ 6 & 8 & 6 & 7^\nu \\ 7^\nu & 6^\nu & 8 & 5^\nu \\ 6 & 7^\nu & 6 & 8 \end{pmatrix}$$

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